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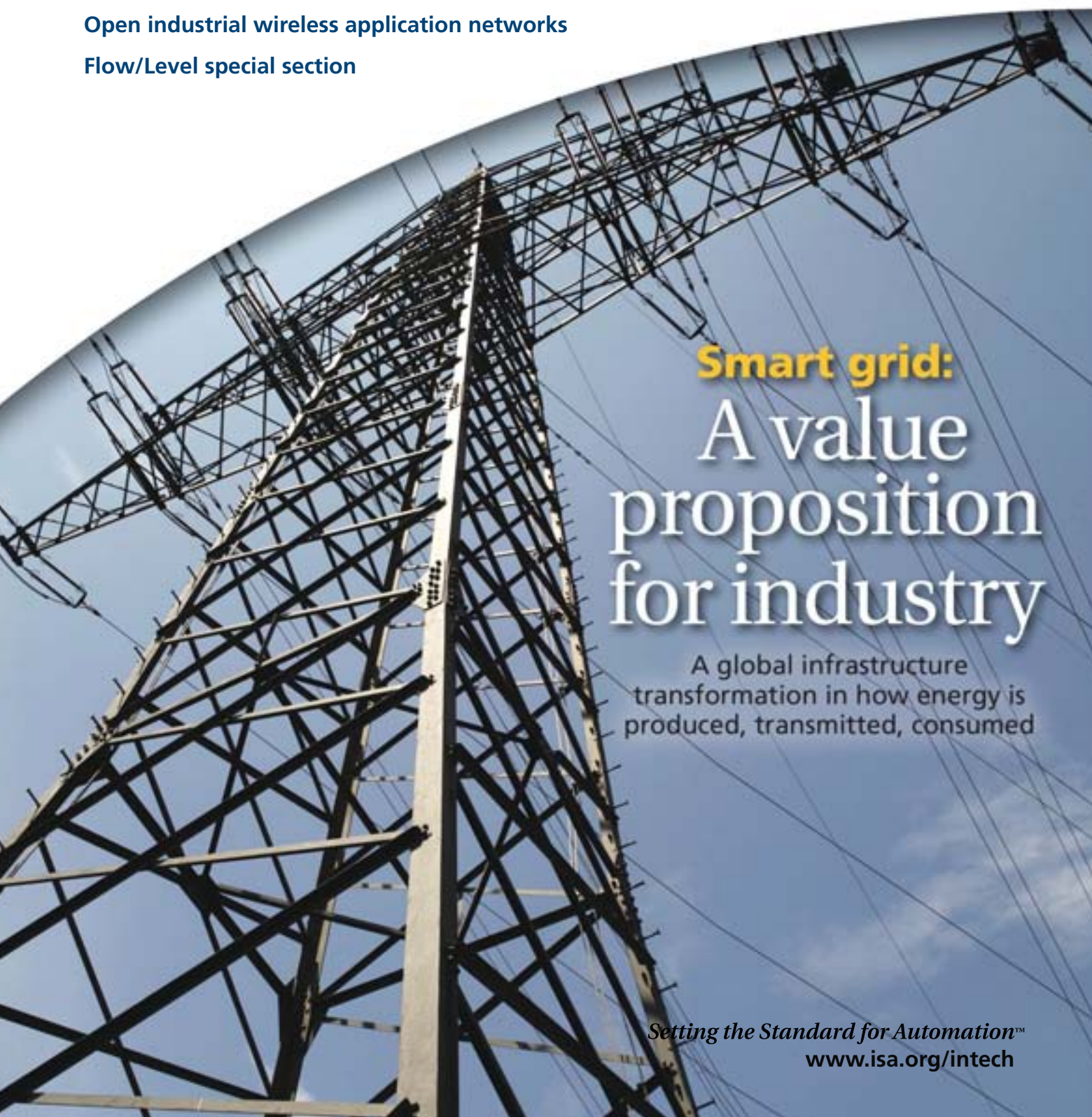
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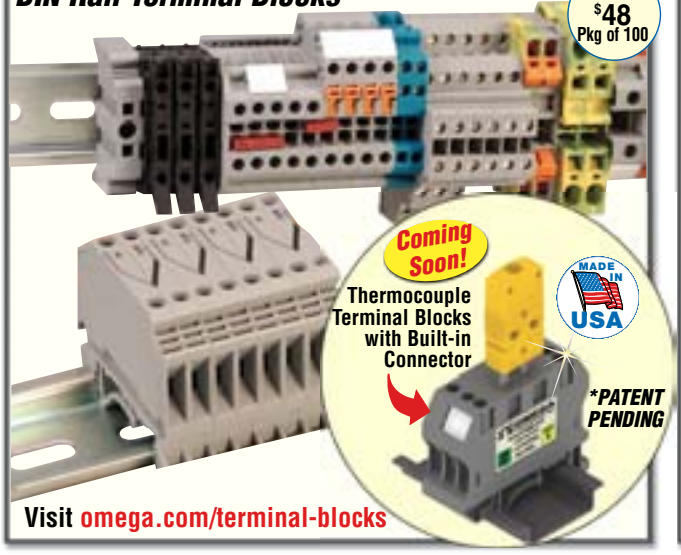
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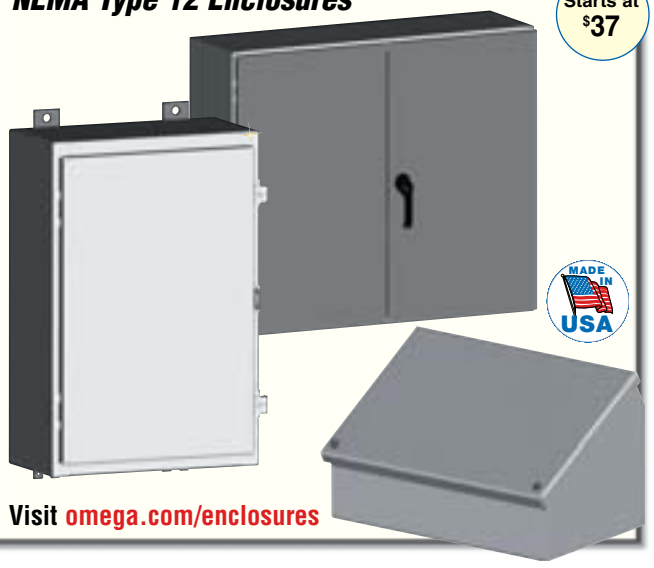
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WEB EXCLUSIVE FEATURES

The human side of safety



"Safety is our first priority"—most companies will agree with this statement, but will recognize it as value of their corporate culture. Indeed, a great deal tention and effort has gone into process safety occupational safety, but one can argue that insufficient has been given to the human aspect of process safety, times called the "human in the loop." Read more at **www.isa.org/intech/201002web1.**

SCADA goes long

SCADA uses long distance communication tools to allow one operator to monitor and control multiple processes spread across miles of countryside, and it does so very effectively. But be careful what functions you put on the communications link. Read more at **www.isa.org/intech/201002web2.**

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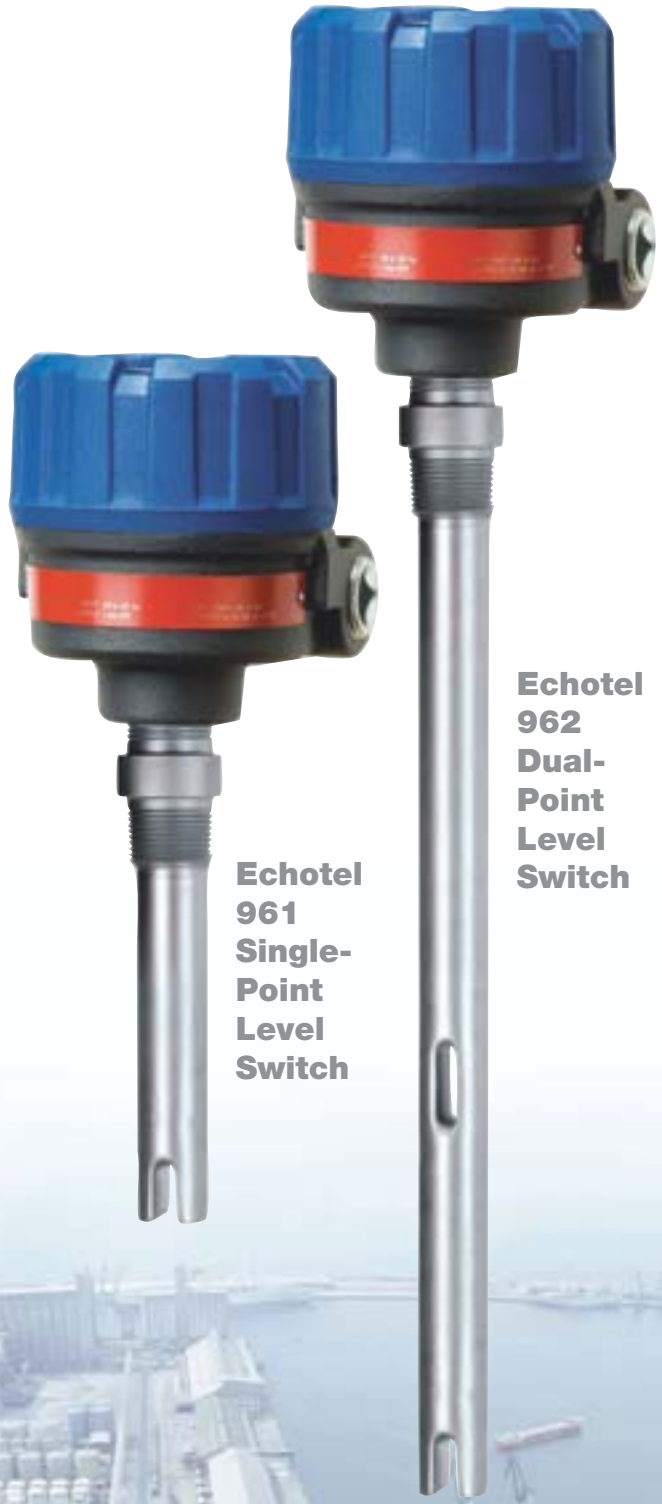
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Be a source of your company's success



By Bill Lydon, *InTech,* Chief Editor

Automation professionals are worth their weight in gold to employers when they apply knowledge and creativity to improving production. Given the current global economic conditions, most companies are dealing with budget constraints, and this situation calls for more creativity and ingenuity. The challenge is to do more with less.

Some ideas I have learned from years of experience, applying Value Engineering, and being trained at the Creative Education Foundation (www.creativeeducationfoundation.org), may be of value.

The power of an idea is underrated, and most people do not spend enough time thinking. Look at what is going on in your production processes, identify what is slowing production, and then come up with a solution. It helps to brainstorm with other people involved, including operators and maintenance people.

Questions lead to creativity, and answers lead to innovations. Make lists of areas needing improvement, and then ask this powerful question for each area: In what ways might we (the challenge) … ?

Make a long list of ideas without worrying about cost or practicality. Include ideas that are silly or impossible because in many cases these ideas open your mind to new practical solutions.

A big obstacle to new ideas is making assumptions such as, "it's never been done that way before," and other negative responses. These thoughts will stop you dead. Here are some things that were not thought possible in the past:

- The first successful cast-iron plow invented in the U.S. in 1797 was rejected by New Jersey farmers under the theory that cast iron poisoned the land and stimulated growth of weeds.
- Men insisted that iron ships would not float, that they would damage more easily than wooden ships when grounding, that it would be difficult to preserve the iron bottoms from rust, and that iron would deflect the compass.

These are some techniques to employ when looking for new ideas:

Rearrange: Making the new by rearranging the old can many times create a better result.

Substitute: What can I substitute to make an improvement? What if I swap this for that and see what happens?

Combine: What materials, features, processes, people, products, or components can I combine? Where can I build synergy?

Adapt: What part of the process could I change? And in exchange for what? What if I were to change the characteristics of a component?

Modify: What happens if I minimize or exaggerate a feature or component? What will happen if I modify the process in some way?

Eliminate: Think of what might happen if you eliminated various parts of the process; this often leads to considering different approaches. What would happen if I removed a component or part of it? How else would I achieve the solution without the normal way of doing it?

Wish: Fantasize how this process could be done if some rules were suspended such as budget, physics, and gravity.

Ideal: What would my ideal solution look like?

It takes courage to explore and create new ideas, but this is how progress is made. Generating many ideas, including impractical, silly, and impossible ones, will open a door to better ideas.

A major function of *InTech* is to enhance the automation profession by communicating and explaining ideas, technology, solutions, and showcasing successes. This information provides everyone in the automation community more ideas to innovate.

As you become a source of your company's success, please share your ideas so we can communicate them to others.

Talk to me at blydon@isa.org.

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The importance of a STEM workforce

I very much enjoyed the Workforce Development department (July *InTech)*, "Engineering field has work to do." Students and parents also suffer under the misguided notion that majoring in a Science, Technology, Engineering, and Mathematics (STEM) discipline prepares graduates only for a career as a scientist or an engineer. The reality is 86% of jobs in high growth industries over the next decade will require postsecondary education, and over 50% require a college degree in a STEM field. Additionally, even traditional non-science jobs will require a more robust understanding of basic science and technology, so people educated in any major will need to have a stronger grounding in STEM fields.

The statistics you note highlight that what is lacking in the U.S. is an appreciation of science and technology and its relevance to peoples' lives. The nation may need only so many Ph.D. educated research scientists and engineers, but there are really good, high-paying jobs in technology fields that go unfilled in this country because employers are having a difficult time finding well-prepared STEM-educated students to fill those positions as Bill Gates notes in several speeches and writings. And, the upcoming wave of retiring baby-boomers educated in STEM fields in the 1960s is lapping at the employment shore.

I am president of the only STEM-focused comprehensive university located between Philadelphia and Pittsburgh in Pennsylvania. When Harrisburg University of Science and Technology was chartered in January 2005, it became the first of its kind in the Commonwealth in more than 100 years. Pennsylvania, a state often seen by many as behind the times, is now positioned nicely on the forefront of higher learning and strategic workforce development.

Although the idea of starting a university in a once predominantly blue collar city was at first greeted with skepticism by many in the community, Harrisburg University has grown a strong philanthropic base of supporters committed to investing in the school and the future prosperity of the region. We expect to exceed our initial campaign goal of raising $40 million.

In 1981, Harrisburg was listed as the second most distressed city in the nation, but in the past 27 years, it has positioned itself as an economic engine, lifting itself from post-industrial economic decline to knowledge-based producer. The capstone to the $2.65 billion in new investments made within the city since those dark days is Harrisburg University of Science and Technology. The University was founded in 2001 by a corps of committed business leaders to transform the state's capital city. They wanted to once again make it a desirable place to live and work as well as drive the region's economic development. They concluded this would happen only if Harrisburg had a university.

In 2000, while 83% of Harrisburg area residents had a high school degree, just 23% of them held a bachelor degree, and only 12% of those living in the region's cities had graduated from college.

Between 1970 and 2000, according to the U.S. Census, the region had lost more than 20% of its manufacturing jobs, while retail and service sector jobs had grown by 83% and 199%, respectively. The region had to find new strategies in order to compete in the knowledge-based economy.

At the time, Central Pennsylvania had 20, four-year colleges and universities that awarded 6,000 degrees annually to a population of 1.7 million. But less than one-third of those graduates had earned degrees in the science and technology fields that the region's businesses needed to make it a major participant in the knowledge-based economy. The idea that innovation will stay here if it starts here was a driving force in the creation of Harrisburg University of Science and Technology.

The public needs to wake up to the importance of a having a well-prepared STEM work force and how it impacts most jobs. We must continue to educate today's students in these important areas and raise awareness of the great careers that are available to people with strong STEM skills.

Dr. Mel Schiavelli
President, Harrisburg University of Science and Technology

St. Clair held in high regard

I enjoyed David W. St. Clair's article on Ziegler and Nichols (July's "The Final Say"). It pleased me to find out St. Clair actually had the opportunity to work with and to know these men.

Having started my career in the late 1980s, I had have held the opinion that those who cut their teeth in controls during the 1940s through 1960s had a much deeper understanding of control theory. I know of only one engineer, who worked for Chrysler during the development of the Saturn rockets, who actually practiced frequency response analysis.

I was given a paper copy of St. Clair's tuning handbook by my first supervisor. This was in the days before the Internet and e-mail, but unfortunately, after the advent of the thermal fax machine. As that print faded, I clung to that article, not realizing his name would one day be so well recognized. As I used that article to better understand the Z/N method (specifically the open loop method; really handy for those two-hour cycle times), I always reached for it when working with a junior engineers who, like I once was, were trying to get a better understanding of efficient tuning techniques. I can't count how many times I have recommended the handbook to younger engineers.

I guess we all go through the basics in linear systems and control theory in college, but for those of us whose careers are centered on chemical plants and refineries, the PID controller becomes one more function block to be configured. I have often wished for the time to take one loop, which is actually in service, and break it down into its characteristic equation, and use those techniques we learned in school, just to prove I CAN do it.

As I recall, the Z/N method was one paragraph in our controls text in college, and I only remembered it then because my professor had mentioned it in a sidebar discussion of what a practicing control system engineer could expect to see on the job.

It seems many engineers and maintenance techs that I have worked with tune by guessing, copying tuning from a similar loop (then guessing), or by using tuning software. I am grateful I was exposed to St. Clair's handbook, as it allows me to explain the method to the younger guys, so they have a better understanding of the subject.

He should realize that guys who came along when I did, hold St. Clair in that same high regard that he holds Misters Ziegler and Nichols.

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Bend, twist, fold: Antenna won't break

Engineers at North Carolina State University have created a highly efficient, flexible, and self-healing antenna using a metal alloy that is a liquid at room temperature.

The new liquid-metal antenna could make it easier to send and receive data from flexible electronics, reported *Technology Review.* Possible uses include sensors incorporated into clothing or other textiles, pliant electronic paper, or implantable biomedical devices.

Michael Dickey, an assistant professor of chemical and biomolecular engineering at NC State, was working with a gallium-indium alloy, which is liquid at room temperature, researching how it behaves in microchannels with a view to electronics fabrication applications. Hunting for other possible uses, he hit on the idea of making a flexible antenna. In collaboration with electrical engineer Gianluca Lazzi—then at NC State, now chair of the department of electrical and computer engineering at the University of Utah—Dickey and his students used the alloy and a common flexible polymer called polydimethylsiloxane (PDMS) to make a simple dipole antenna—essentially a straight rod, like the old-fashioned "bunny ear" antennas used for analog TV.

The researchers poured liquid PDMS into a mold that left it with a single internal channel once cured. They then injected the liquid gallium-indium mixture into the channel and sealed it.

Researchers at Lazzi's lab tested the antenna's performance and found they could create an electrical contact with the device simply by jabbing a wire into the liquid, eliminating the need for solder. In the lab, the antenna radiated over a broad frequency range at about 90% efficiency. The antenna also remained functional while the engineers bent, twisted, and folded it in half; they even stretched it an additional 40% beyond its normal length. When the stress was released, the PDMS snapped back to its original shape.



A flexible antenna consists of liquid metal injected into microchannels in a stretchy polymer. Source: North Carolina State University

Garbage fuel for garbage trucks

Hundreds of trash trucks across California are rumbling down city streets using clean fuel made from a dirty source: garbage.

According to *The Associated Press,* the fuel is derived from rotting refuse San Francisco and Oakland residents and businesses have been discarding in the Altamont landfill since 1980. Since November 2009, the methane gas created from decaying detritus at the 240-acre (96-hectare) landfill has been sucked into tubes and sent into an innovative facility that purifies and transforms it into liquefied natural gas.

Almost 500 Waste Management Inc. garbage and recycling trucks run on this new source of environmentally friendly fuel instead of dirty diesel.

In a state that has passed the most stringent greenhouse gas reduction goals in the U.S., the climate change benefits of this plant are twofold—methane from the trash heap is captured before entering the environment and use of the fuel produces less carbon dioxide than conventional gasoline.

"We've built the largest landfill-to-LNG plant in the world; this plant produces 13,000 gallons (49,400 liters) a day of LNG," said Jessica Jones, a landfill manager for Houston-based Waste Management. "It will take 30,000 tons a year of CO_2 from the environment."

Altamont is one of two California landfills making LNG; the other is a smaller facility about 40 miles (65 kilometers) south of Los Angeles. Other natural gas facilities are being planned by Waste Management at some of the 270 active landfills nationwide, and the number could grow quickly as communities seek to reduce greenhouse gas pollution.

More full-body scanners coming to airports near you

Since explosive materials were sneaked onto a U.S.-bound flight from Amsterdam on Christmas Day 2009, full-body scanning machines are more likely to make their way to security lines at your local airport, even though they might not have detected said materials.

While the Transportation Security Administration already has 40 such devices in place, it bought 150 in January to be placed in U.S. airports and said it plans to buy 300 more (they go for $170,000 apiece).

These full-body scanners fall into two main categories: millimeter wave and backscatter, according to *CNET*. The first directs radio waves over a body and measures the energy reflected back to render a 3D image. The latter is a low-level X-ray machine that creates 2D images.

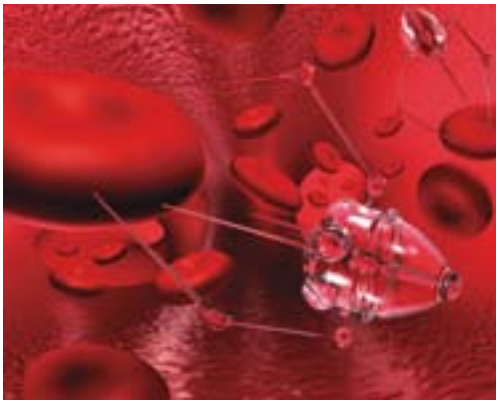
The scanners are supposed to be the high-tech (and energy-inefficient) version of a pat down, and can detect items such as nonmetallic weapons and explosives not picked up by metal detectors.

Millimeter wave scanners produce 30 to 300 gigahertz electromagnetic waves, and reveal explosives if they are denser than other materials. This means these scanners emit less radiation than a typical cell phone, according to TSA.

The backscatter machines are low-level X-ray machines that expose bodies to as much radiation as about two minutes of flying in an airplane does. In other words, if you use a cell phone and you fly, you are already exposing your body to more radiation than these scanners will.

Automation by the Numbers

30/40

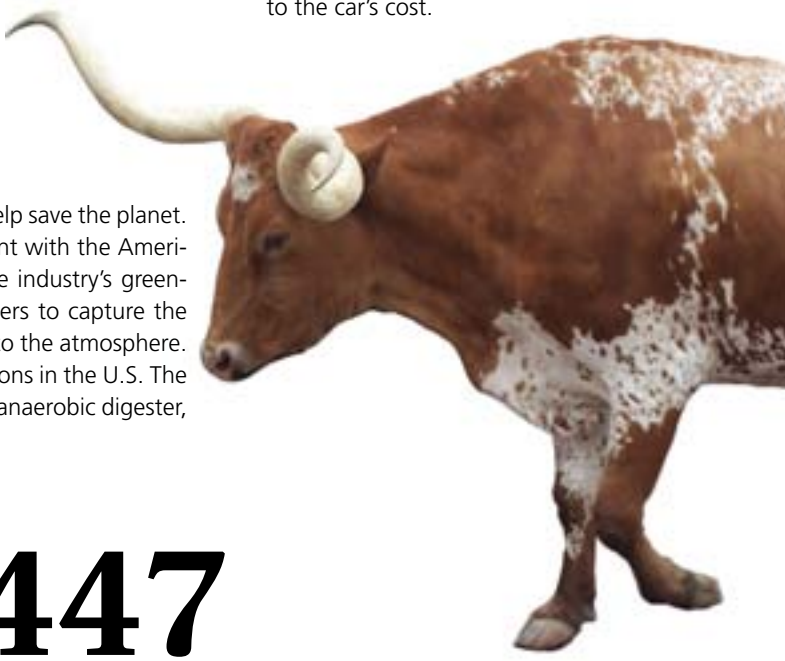


In **30 or 40** years, we will have microscopic machines traveling through our bodies, repairing damaged cells and organs, effectively wiping out diseases. Author and futurist Ray Kurzweil said anyone alive come 2040 or 2050 could be close to immortal, according to his interview with *Computerworld*. The quickening advance of nanotechnology means the human condition will shift into more of a collaboration of man and machine, as nanobots flow through human blood streams and eventually even replace biological blood, he added. Limbs could be regrown. Backed up memories and personalities also could be accessed after a head trauma.

2028

The National Research Council issued a report that estimates it could be **2028** or later before the fuel savings outweigh the additional up-front cost for plug-in vehicles. The report said the biggest reason for the higher up-front cost of a plug-in is the battery. The battery pack for a car capable of going all-electric for 10 miles would add about $3,300 to the cost, the authors estimated. They said the battery pack for a car that can go 40 miles without using gas would add about $14,000 to the car's cost.

25%



The U.S. is counting on cows to help save the planet. *The Associated Press* reported an agreement with the American dairy industry is in place to reduce the industry's greenhouse gas emissions **25%** by 2020, mostly by convincing farmers to capture the methane from cow manure that otherwise would be released into the atmosphere. Agriculture accounts for about 7% of the greenhouse gas emissions in the U.S. The plan calls for persuading more American farmers to purchase an anaerobic digester, which essentially converts cow manure into electricity.



-447

NASA's Wide-field Infrared Survey Explorer (WISE) lifted off over the Pacific Ocean at the end of 2009 on its way to map the entire sky in infrared light. WISE will see the infrared colors of the whole sky with sensitivity and resolution far better than the last infrared sky survey, performed 26 years ago. The space telescope will spend nine months scanning the sky once, then one-half the sky a second time. The primary mission will end when WISE's frozen hydrogen runs out, about 10 months after launch. Because the instrument sees the infrared, or heat, signatures of objects, it must be kept at chilly temperatures. Its coldest detectors are less than **-447°F.** Near-Earth asteroids, stars, planet-forming disks and distant galaxies all will be easy for the mission to see. Hundreds of millions of objects will populate the WISE atlas, providing astronomers and other space missions with a long-lasting infrared roadmap.



Smart grid:

A value proposition for industry

By Dave Hardin

Sustainability and energy management are more than green issues driven by social responsibility. They are becoming economic imperatives in industry and manufacturing. "Smart grid" can help achieve both objectives.

Manufacturing is trying to recover from one of the deepest downturns in recent history. Consumer spending is down, manufacturing plants have closed, and personnel reductions are widespread. Companies are in survival mode and struggling to control expenses. Many are relocating operations to lower cost regions, all the while slashing capital budgets. This is not a time of prosperity for manufacturing and service industries as our economy undergoes major changes. While signs of life conjure up optimism, it is clearly a time for retrenchment.

FAST FORWARD

- In the U.S., electrical energy use in the industrial sector is about 25% of the grid's total energy.
- Electrical disturbances on the grid can affect power quality and reliability.
- Industrial microgrids are self-contained energy systems that include the capability to consume on-site energy generation as well as grid-supplied generation.

Global infrastructure transformation in how energy is produced, transmitted, consumed

Meanwhile, the media is alive and well with stories about the green jobs being created, the need to reduce our carbon footprint, the deployment of smart electric meters to homes throughout the country, a new wave of plug-in electric vehicles getting ready for production, cyber-security vulnerabilities of our power grid, and the threat of global warming unless we get serious about our environment. Fueling the media coverage, the U.S. government issued a statement that billions of stimulus dollars will be provided to kick start the repowering of the U.S. energy system with what is called the "smart grid." And this is just the beginning.

While amounts vary, The Brattle Group, which provides consulting and testimony in economics, finance, and regulation, estimates the total investment in the U.S. energy system could reach as high as $1.5 trillion dollars over the next 20 years. Something big is going on!

At first glance, it does not seem like something that industry should pay much attention to. New transmission lines, distribution systems, electric cars, and smart meters in homes are all well and good, but they have not been very interesting to industries such as manufacturing. Maybe it is time to reconsider involvement.

Impact on the grid

The Energy Information Administration of the U.S. government in the report "International Energy Outlook 2009" stated: "Energy is consumed in the industrial sector by a diverse group of industries—including manufacturing, agriculture, mining, and construction—and for a wide range of activities, such as processing and assembly, space conditioning, and lighting. … In aggregate, the industrial sector uses more energy than any other end-use sector, consuming about one-half of the world's total delivered energy."

In the U.S., electrical energy use in the industrial sector is about 25% of the grid's total energy. This is down from 33% in 1996 and reflects the decline in U.S. industrial capacity. In spite of this trend, electrical energy is still a major operating expense of many industrial operations. The top 10 electrical consumer groups in industry are:

1. Chemicals
2. Primary Metals
3. Nonmetallic Minerals
4. Paper
5. Non-ferrous Metals
6. Food
7. Plastics and Rubber Products
8. Transportation Equipment
9. Computer and Electronic Products
10. Textile Mills

Chemicals, primary metals, minerals, and paper constitute more than 60% of the industrial electrical consumption. These energy intensive industry sectors are the most sensitive to variation in energy costs.

Industry also plays a role as a producer of electrical energy. The contribution of electrical energy generation from industrial sources is about 4%. While this may seem insignificant, it is equivalent to current non-hydro renewable energy generation.

Impact of the grid

It is clear industrial consumption has a significant impact on the grid; however, it also works the other way. The electrical grid has a major impact on industrial customers.

Electrical disturbances on the grid can affect power quality and reliability. Bulk power plant fail-

ures, transmission congestion, stresses caused by peak demand, and distribution failures all contribute to conditions that can increase operational costs and curtail productivity. Industrial energy efficiency suffers due to the above-normal energy required to resume operating pressures, temperatures, and momentum after each electrical disturbance.

A Lawrence Berkeley National Laboratory report estimates electric power outages and blackouts cost the nation about $80 billion annually. Of this, $20 billion represent losses to the industrial sector and $57 billion to the commercial sector. This is misleading, as the impact to industrial customers in real terms is significantly greater than the impact to commercial customers.



The industrial sector uses about one-half of the world's total delivered energy.

Another impact is that of power quality where voltage surges and sags can affect electronic equipment operation unless protective equipment and backup power generation have been installed at extra cost.

And cost is the key. Cheap energy has fueled growth, but cheap energy is becoming not-so-cheap. Average retail costs have been on a consistently upward trend, even as consumption has stabilized during the recession.

Infrastructure transformation

The smart grid represents a global infrastructure transformation in how energy is produced, transmitted, and consumed, driven by the capabilities of modern automation, communication, and information technology. According to the Energy Independence and Security Act 2007, Title XIII, this transformation will include:

- Integration of distributed renewable energy generation
- More reliable transmission and distribution
- Bi-directional energy flow with customers
- Intelligent energy consumption.

Critical first steps have already been taken, but the path to a smarter grid is a journey, not a destination. Initial funding under the American Recovery and Reinvestment Act includes $4 billion for smart grid investment grants and demonstration projects. This is in addition to $42 billion for energy efficiency and renewal energy programs and $21 billion in energy incentives. It is important to realize this funding represents only a fraction of the total investment required.

Driving change

The transition to a smarter grid will drive change. No longer will energy be a bill that shows up a month after it is used. Instead, energy usage and cost information will be available for timely analysis and financial decision making. Energy will be a cost that is dynamically controllable. While this is currently the case with some large industrial customers, dynamic energy management will become widespread.

Energy price and demand shaping will be used to make financial decisions not only related to energy consumption but also to energy production. If the price of energy is high then producing power locally and selling back to the grid may make good financial sense. When the price is low, industrial processes that are financially marginal may become economic to operate. This decision making will require intelligent automation systems but …

With change comes opportunity

Industry should embrace smart grid as an opportunity to blend sustainability with profitability. Not only should companies focus on internal projects that reuse and optimize energy, but they should evaluate the impact they can have on their community. Industrial companies can capitalize on their larger size and rural locations to become net energy exporters in addition to maintaining efficient operations. Opportunities come in many flavors.

Energy management

Energy is a valuable resource that needs to be managed. The first and most important step is to understand where and how energy is being consumed or exchanged and the impact of that consumption on operational economics. Energy management systems can help by providing transparency into energy usage through key economic indicators that can aid in the decision-making process, both in offline design and online optimization. Stabilization and demand shaping can improve energy efficiency by as much as 10%, which for some heavy process facilities represents more than 30% of their operating profits.

Many industrial processes are decades old, dating from a time when energy was cheap and abundant. These represent areas of opportunity for creative energy management solutions.

An important consideration is the ability to upgrade on-site energy systems to enable integration with smart grid signals such as dynamic pricing, curtailment demand response, and reliability signaling.

Industrial microgrids

Industrial microgrids are self-contained energy systems that include the capability to consume on-site energy generation as well as grid-supplied generation. This flexibility provides insulation from grid faults that affect power availability and quality while also permitting excess energy to be exported back to the grid. The decision to import or export energy is controlled by an energy management system that optimizes consumption with generation.

In many cases, local backup generation is already required due to the negative operational impact of an electrical failure. If the backup generation capac-

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Industrial sites may be well-suited for wind farms.

ity is significant, then some of the core elements of a microgrid may already be in place.

Distributed renewable generation

Industrial facilities are often built on large plots of land and are located in areas conducive for the installation of renewable power generation, such as wind, solar, geothermal, and biofuels, on the distribution grid. In this case, generation is controlled directly by the local electrical utility or aggregator with benefits accruing through land lease and other agreements.

Many industrial facilities currently operate gas-fired or coal-fired co-generation (combined heat power) power plants, and many of these are importers and exporters of grid power. A financial opportunity may exist to leverage and expand upon these existing installations.

Industrial control of variable energy requires advanced automation and optimization. There exists an opportunity within the automation industry to adapt existing modeling and simulation technology toward this important application.

Bulk renewable generation

In the case where an organization has sufficient real estate and financial resources, the development of a bulk generation station based on diverse renewable resources that are geographically-dispersed may be an option that makes fiscal sense. Solar energy during the day can be balanced by wind energy at night with gas turbines and storage providing energy balance when needed. This would permit a virtual power plant to be operated as if it were a single "dispatchable" generator. "Dispatchable" resources provide grid operators with known generation capacity that can be brought online and taken offline as needs arise on the grid. Solar and wind generally provide "base load" generation.

A long way to go

The ideas presented could be impacted greatly by current events. The United Nations held its Climate Change Conference in Denmark at the end of 2009, with cap-and-trade being debated on Capitol Hill in Washington, D.C. Limits could be imposed on greenhouse gas emissions impacting industrial facilities worldwide and bending the economics further toward the concepts presented. Even if little results from these global initiatives, there is still a high probability the true costs of energy in terms of security and environmental impact will get factored into the price.

The qualitative affect of green house emissions is understood and widely accepted even as the quantitative impact is being modeled and debated.

World peak oil production is also vigorously debated, but few estimates reach beyond 2030. A recent article in *The Economist* quotes Faith Birol, chief economist of the International Energy Agency, saying she "believes that if no big new discoveries are made, 'the output of conventional oil will peak in 2020 if oil demand grows on a business-as-usual basis.' " Many of us will personally experience the negative effects of peak oil as it impacts the global oil-dependent economies.

Solar and wind have the potential to replace a very significant percentage of our oil consumption in the long run. This will require new technology and systems that transform solar and wind farms into first-class generation resources. In addition, coal-fired plants using CO_2 sequestration are poised to be cost competitive with wind providing improved predictability. One thing is for sure, the world will need energy from many sources, all integrated on a smarter grid and working together.

Call to action

Industrial customers have the opportunity to be active and effective participants in transitioning toward a smarter grid. The new regulations, business models, technology, and communication standards required are in the process of being formulated, and their effect on industry will be significant.

Organizations have two fundamental options: 1) passively wait and accept what happens or 2) get involved and help shape smart grid regulations and standards through active engagement. The National Institute of Standards and Technology (NIST) welcomes industrial user participation in the standards process. If interested, e-mail Keith Stouffer at keith.stouffer@nist.gov. More information can be found at http://www.nist.gov/smartgrid.

Understanding the impact of energy on your business operations and developing a proactive, long-term business strategy that recognizes and leverages smart grid opportunities will be investing in the business's future, the country's future, and the world's future.

ABOUT THE AUTHOR

Dave Hardin (David.hardin@ips.invensys.com), Invensys Operations Management, serves as a member of the Department of Energy-supported GridWise Architecture Council and co-chair of the NIST Industrial-to-Grid Domain Expert Working Group. Dave also serves as a member on the NERC Smart Grid Task Force and OPC Foundation's Technical Advisory Council. He is a Registered Professional Engineer (P.E.) and Project Management Professional (PMP).

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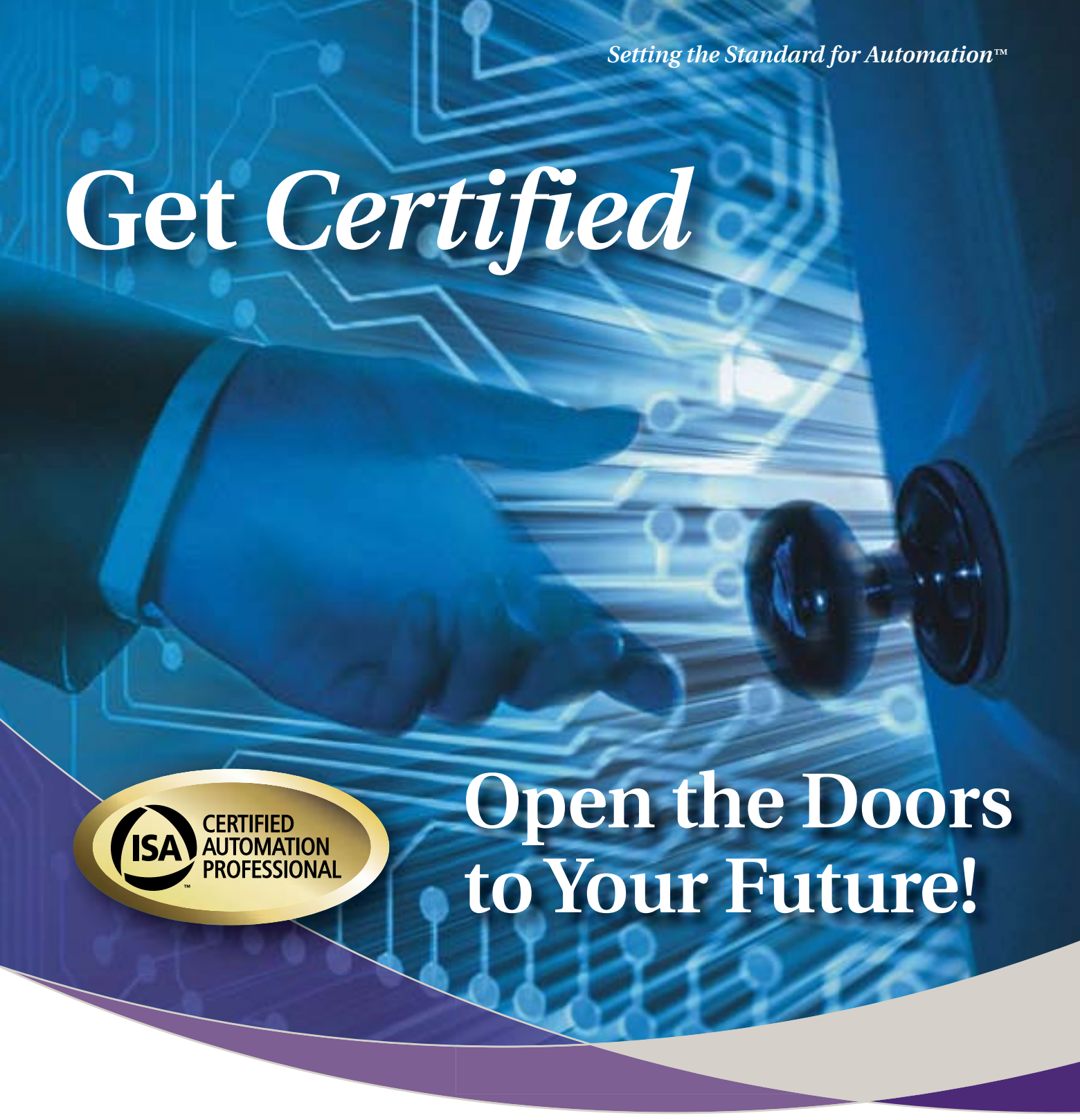
U.S. Energy Information Administration
www.eia.doe.gov

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http://certs.lbl.gov/certs.html

Wireless and the Smart Grid
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Without a trace

Enterprise integration key to track and trace success

By Dennis Brandl

Our food supply chain is large, complex, and diverse, and traceability in food and pharmaceutical products is a life-critical issue. Counterfeit drugs containing little if any active ingredients, counterfeit raw materials found in pharmaceutical manufacturing, contaminated peanuts, peppers, and meat have all led to sickness and death. The food industry needs an over-arching, global food traceability system that can link all food products and provide results in seconds. But the problem of traceability does not stop at the plant door. In order to meet the real requirements for traceability and recall control, some sort of method is required for cross-enterprise traceability.



Traceability facts

Traceability means producers must keep track of where they obtained their raw materials and where they shipped their product. This is complicated because lots split and combine during production. The end product on a store shelf probably came from several different raw material lots from several different sources. A single container of mixed-berry yogurt might have fruit from two sources, milk from a third source, and culture from a fourth source. A packaged meal might have meat from one source and vegetables and fruit from multiple sources. The supplier might have combined the vegetables and fruits from different suppliers. Even the municipal water being used in production should have an identifying lot number associated with the day or even shift of use. Complete traceability might require multiple companies to combine their traceability information and share lot numbers across the entire supply chain.

Within a plant, the main issue of traceability is to determine exactly which lots went into the final product. While lot assignments are the norm in drug manufacturing, food and beverage production schedules do not specify which raw material lots to use. It is at the discretion of operators to pick the appropriate material lot for each batch. In-plant traceability requires identification of ingredients as they are added and identification of final products at lot boundaries.

Labels, packing

Packing and labeling is where track and trace has the biggest impact. The packing line is where manufacturers label products, add RFID tags, apply bar codes, and add other external tracking information. As more products require tracking and traceability, including food, tires, electronics, car parts, pharmaceuticals, consumer products, and even toys, additional demands fall on packaging equipment. This usually requires automation of the packaging lines to individually print labels and program RFID tags. It also requires integration with corporate systems to download tracing information and upload manufacturing information associated with the items.

The old days of putting a product in a box, slapping on a label, and sending it to shipping does not work anymore. A product's packaging is an integral part of a product specification. The package provides external traceability through the supply chain, speeding recalls and replacement. Packaging also calls for intelligent routing of products within supply centers to maintain shipping schedules and minimal distribution costs.

Global efforts

Current government regulations do not ensure quick traversing of links through the supply chain, making a time-consuming task of determining source issues in the event of a contamination. Some vertical tracking systems are starting to emerge within segments of the industry, but once cross-segment components mix (putting strawberries in yogurt or eggs in cookies), the links can once again become time-consuming to trace.

Without accurate information regarding which products might be impacted in a contamination concern, recalls fall on the side of caution and safety, severely impacting entire product types, which are later found to be unrelated to the actual issue.

Efforts by World Food Trace, OpenO&M, and ISA95 to provide cross-enterprise integration through world-wide traceability are critical in the food and pharmaceutical manufacturing of today. Such efforts rely on internally integrated business and production systems to generate genealogy data. They also need evolving standards for the cross-enterprise exchanges worldwide that tracking and tracing require.

Cross-enterprise traceability

Cross-enterprise traceability requires assignment of globally unique IDs to individual lots, similar to globally unique IDs assigned to phone numbers, Internet addresses, and vehicle identification numbers. Current standardized ID methods deal with identifying the product, not the specific lots.

Traceability of the components of these products is now the law of the land in quite a few countries. The U.S. has the Bioterrorism Act, Public Law 107-188, and the EU has the European Health and Consumer Protection Directorate 178/2002. These regulations allow government agencies to quickly address contamination issues and other food scares and to mitigate bioterrorism attacks.

The basic concept: It should be possible to identify every ingredient in any food and trace the ingredients back to its source. This allows for faster and more focused recalls, less risk to the general public, and less risk to manufacturers.

In addition to bioterrorism and food safety issues, several countries have developed mandatory labeling laws for foods containing ingredients derived from genetically modified (GM) crops. To comply with these labeling laws, food manufacturers must be able to document the genetic purity of GM and non-GM ingredients. They can do this either by preserving the identity of a crop from seed to final product or by tracing from the final product to the crops from which ingredients were manufactured.

FAST FORWARD

- Deadly food products call for traceability action in industry.
- Global industry efforts moving ahead, but not enough.
- Cross-enterprise traceability method needed for real change.

While the current U.S. Bioterrorism Act requires traceability known as "one up and one down," recent food scares such as contaminated peanuts and peppers have demonstrated true recalls and safety announcements will require multiple producer cooperation.

Past recalls

In 2008, from April to June, a salmonella outbreak sickened over 150 people and caused the Food and Drug Administration (FDA) to recall tomatoes and fresh tomato products on 3 June. A week later, authorities cleared the tomatoes as the cause, but still had no identified source. Finally, on 21 July, the FDA announced jalapeno peppers grown in Mexico were the cause. In the end, 1442 people from 40 states became ill, and falsely implicated tomato growers lost an estimated $450 million after 12 weeks had gone by. This incident illustrates how our ability to quickly and correctly identify the source of food-borne contamination needs improvement.

In November and December 2008, a salmonella outbreak led to eight deaths, sickened over 600 people, and caused Peanut Corporation of America (PCA) to recall its products. Six months later, products were still being identified as containing peanuts from PCA. With the total of over 3,200 products, 470 separate company recall notices, and the cost of millions to companies, nobody was assured the search to find all uses of PCA peanuts in the food supply chain was complete.

An AMR study found over 50% of food companies participated in health and safety recalls within the past year. It took an average of 14 days to sense the need for the recall and over 30 days to enact it. Authorities could actually recall less than 40% of the product, and the financial impact of the recalls totaled hundreds of millions of dollars. Companies who did their best at recalls had multiple-enterprise tracking, manufacturing automation, mature supply chains, lot-level tracking, and cross-functional team reviews of quality.

Pharmaceutical manufacturers have similar

traceability problems. In April 2009 in Korea, patients died after taking drugs made with talc that contained asbestos from China. In 2008, there were deaths due to heparin contamination.

The 2006 Compliance Policy Guide for the Prescription Drug Marketing Act refers to a drug pedigree as a "statement of origin that identifies each prior sale, purchase, or trade of a drug, including the date of those transactions and the names and addresses of all parties to them." An e-pedigree is an electronic document that satisfies a pedigree requirement. Its main purpose is to protect consumers from contaminated medicine or counterfeit drugs.

Industry mitigation efforts

In the pharmaceutical and biotech industries, much of the effort to address traceability problems lies in companies' internal tracking. But the food industry is establishing cross-functional organizations, some of which deal with specific food products, such as fresh fruit and produce growers and shippers through HarvestMark (www.harvestmark.com) and eProduce (www.eproduce.biz). Similar efforts are in place for direct-from-farm meat products.

One organization is approaching the cross-enterprise traceability issue for

Standard lot identification needed

A variety of lot code markings and systems are prevalent in the industry now, making a singular traceability system unworkable. The industry needs a new universal 16-character code as a consistent method of identification for all food products. The manufacturer should be able to self-generate each code, but still identify the lot's origin with absolute guarantees of no lot duplication. This code should fit into a simple bar code, and must be mathematically proven to last globally for every lot of food produced until at least the year 2100.

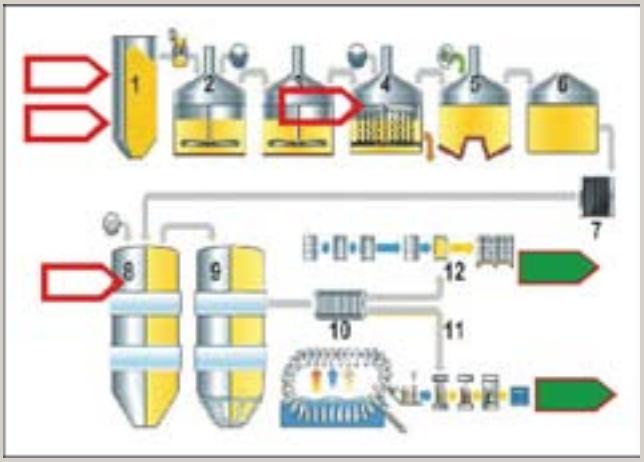
World food IDs can be uniquely keyed with the first seven characters to food processing exit points as registration sites in the food supply chain. This provides over 1 trillion possible production lines. The remaining nine characters form a time/date stamp for lot uniqueness. This provides over 5 trillion possible production lots. A manufacturer with four packaging lines would register four world food IDs, one for each line to uniquely identify the production line as well as the lots coming off the line. At some interval, all produced lot IDs would go to a registered data center for secure and protected storage.

The collection of world food IDs from all registered data centers would form a traceability map of the entire supply network. This allows for forward and backward traceability. Standardized interfaces to the data centers, through secure web services, would provide authorized legal entities access to the entire supply map. These goals require identification of each raw material source and each product exit point.

When a product generates at an exit point, the producer must submit a product lot record to a registered data center. In recording the lot, the producer captures the IDs of all the contributing components of his lot and associates them with the new product's ID. This linking to component globally unique IDs is the foundation of the instantaneous traceability within the system.

This system requires an independent non-profit agency to register the data centers. Agencies must provide a set of criteria that licensed information service providers must meet in order to be official licensed data centers. The worldwide community of data centers thus forms a distributed database of global food supply-chain information.

Using commercial data centers instead of a single universal database means a competitive environment for data services



Material entry points are at 1, 2, 4, 5, and 8. Material exit points are at 11 and 12. An ingredient container would be scanned at the five entry points. IDs would be generated at the two exit points and containers labeled.

and allows food producers to choose a data center that fits their needs. Large manufacturers can even obtain a license for their own information technology department as an approved data center. With proper authority and access, agencies can query the global network of data centers to investigate and respond to food contamination concerns at Internet speeds.

More requirements include:

- No collected proprietary food product information, only recorded lot tracking data
- Non-profit registration agency that can work independent of manufacturers, reducing concerns about sharing of potential market information with competitors
- Distributor and producer endorsements, such as the Grocery Manufacturers Association, International Dairy Foods Association, and GS1 US
- Support from national and international standards bodies
- Support from government regulatory agencies
- Use of self-sustaining model that will return value back to the industry as improved food traceability and other food safety initiatives

Source: Dennis Brandl (dnbrandl@brlconsulting.com) is president of BR&L Consulting, Inc. in Cary, N.C.

all produced foods (dairy, fruit, produce, meat, and the like) direct to consumer and through processed food manufacturers. World Food Trace (www.worldfoodtrace.org) is a nonprofit organization that uses registered databases for cross-enterprise traceability.

Some of these organizations are working with industry groups as well as national and international standards organizations. The ISA88 standard committee already has a standard in place for batch production records, which can be a source of traceability data. The ISA95 standard is extended to the batch production record standard to include all aspects of production, including discrete and continuous production, packaging, shipping, and receiving.

Traceability system requirements

The requirements for systems to provide traceability are fairly straightforward:

1. Create a food supply chain traceability system that identifies all source component ingredients of a food product applicable broadly in all industry segments within seconds.
2. Enable forward tracing of any ingredient to identify all associated products within seconds.
3. Ensure a smooth and rapid implementation with recommended practices to aid food industry manufacturers and processors in the application and use of the system.
4. Address public safety concerns regarding the nation's ability to track and trace food products.
5. Provide a traceability system with the lowest possible cost and burden on the industry.
6. Be an asset to industry as a self-sustaining system with the ability to support other food safety initiatives.

Meeting these goals will require a standard lot identifier for all food and food components as well as record-keeping guidelines and guidance criteria for implementers that will allow for small and large entity success. Manufacturers will also need registered data concatenation centers, which hold the traceability information for access by authorized agencies and for individual company traceability actions. A registration mechanism is required for information service providers to ensure global search capability.

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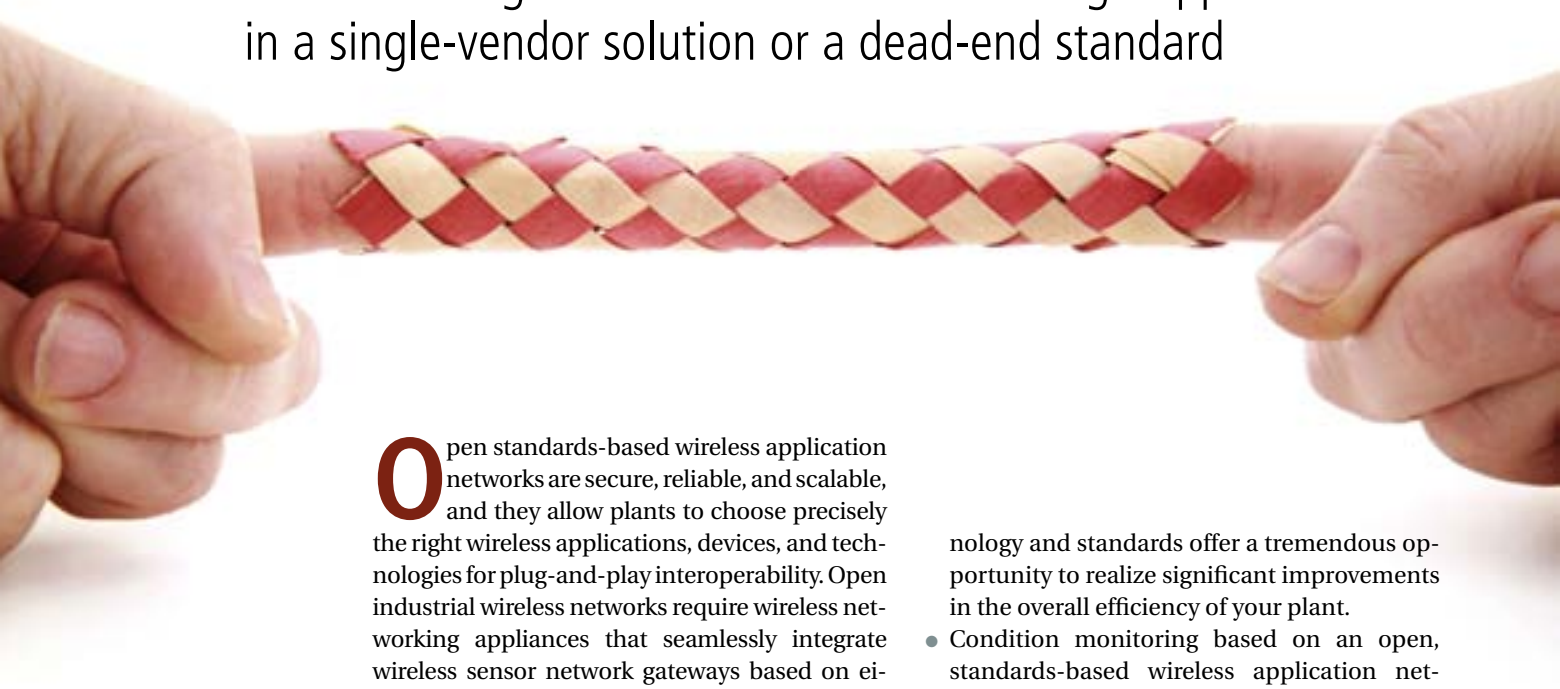


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Open industrial wireless application networks

Integrated support for ISA100 and WirelessHART networks means moving forward without fear of being trapped in a single-vendor solution or a dead-end standard



By Stephen Lambright and Sarah Prinster

Open standards-based wireless application networks are secure, reliable, and scalable, and they allow plants to choose precisely the right wireless applications, devices, and technologies for plug-and-play interoperability. Open industrial wireless networks require wireless networking appliances that seamlessly integrate wireless sensor network gateways based on either ISA100.11a or WirelessHART standards with 802.11 (WiFi) radios to support other industrial wireless applications such as mobility, location, video, and communications and to enable efficient backhaul of sensor data wirelessly. By supporting devices with these standards, industrial facilities will be equipped with the easiest, most cost-effective way to gather accurate and timely information to improve safety, product quality, and productivity plant-wide.

Benefits of an open industrial wireless application network

- An open, standards-based industrial networking appliance and devices offer processing manufacturers the option to choose either ISA100.11a or WirelessHART standards with 802.11 (WiFi) radios.
- Wireless sensor networks can enable better, more timely data into your control system, predictive maintenance, or asset management application.
- The new generation of wireless network technology and standards offer a tremendous opportunity to realize significant improvements in the overall efficiency of your plant.
- Condition monitoring based on an open, standards-based wireless application network enables cost-effective, reliable, and safe instrumentation for critical facility measurements needed to optimize plant efficiency.
- Wireless instruments based on ISA100.11a or WirelessHART deliver data to the full range of maintenance, safety, and security applications.
- Reliable, long-range, high bandwidth wireless broadband technologies like WiMAX (IEEE 802.16) have been ruggedized for industrial environments and applications. These industrial broadband networks are designed to deliver improved reliability and wireless capacity.
- An open, integrated wireless application network enables plants to use many "best of breed" applications for increased ROI and lower total investment costs, which can be sustained for many years.

Wireless technologies at work

One does not have to look far to see major automation vendors are increasingly integrating wireless technologies into their products. Going wireless is seen as a way to cost effectively add

more process monitoring capabilities, enhance workforce mobility, improve safety and security, and drive greater utilization of assets, raw materials, and energy. However, for wireless to work in manufacturing environments, the technologies must deliver reliable performance, security, and ease of use.

Although wireless is heralded as the next big thing in automation, it certainly is not new. The move to use wireless technology to reduce costs and improve efficiency has been underway for some time in manufacturing organizations. What has changed is the emergence of products, applications, and standards to address the specific challenges for using wireless in large manufacturing facilities. By extending the range and lowering the costs of plant and process network communications, this new generation of wireless network technology offers a tremendous opportunity to realize significant improvements in the overall efficiency of the plant.

Wireless sensor networks can enable better, more timely data into your control system, predictive maintenance, or asset management application. Operators in the field are now able to see the control system and review standard operating conditions, procedures, and corrective actions in real time as they make field adjustments. Security departments are using wireless as a means to improve security and achieve timely compliance with increasing regulations by wirelessly adding video monitoring, along with improved access control and intrusion detection. Other technologies and applications such as voice communications and asset tracking use wireless to enable productivity gains that have already been realized in other industries such as healthcare and transportation.

In some cases, these benefits were simply too costly to achieve by running wires, while in others they simply could not be done without wireless networks. Going forward, the challenge is to ensure the best of breed industrial wireless solutions are secure, reliable, scalable, and simple to operate. Overcoming that challenge requires a clear understanding of the alternative technologies and how to best apply them.

Choosing the right wireless technology

As the list of wireless applications grows, so do the number of wireless devices and systems that support these applications. This application growth adds complexity from using multiple wireless technologies to address each application's specific requirements for coverage, latency, and throughput.

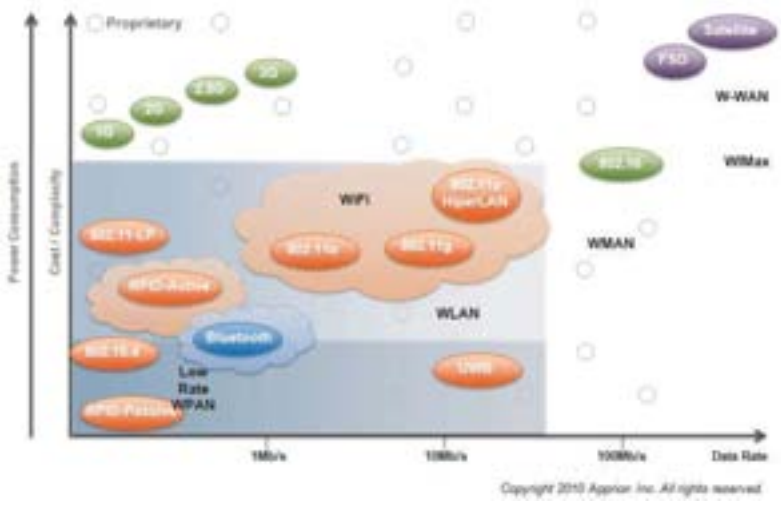
Most wireless systems have been designed to use public frequencies. The frequency sharing is made easier by the emergence of robust standards for communications, but standards alone are not enough. Standards assure the proper function of the systems with a given set of cost/performance characteristics and a basis for interoperability—but no single wireless technology or standard is capable of being the single solution for every application.

FAST FORWARD

- An open, standards-based industrial networking appliance and devices offer processing manufacturers the option to choose either ISA100.11a or WirelessHART standards with 802.11 (WiFi) radios.
- An open, integrated wireless application network enables plants to use many "best of breed" applications for increased ROI and lower total investment costs.

This diversity of cost/performance tradeoffs among the dozens of available industrialized wireless networking technologies dictates users choose the most effective technology and devices for a specific application. There is not a "one-size-fits-all" wireless networking technology that adequately supports the diverse and demanding requirements of the many different kinds of industrial applications and environments.

Standards-based technologies like WiFi (IEEE 802.11) have been hardened for mobile workers to take ruggedized tablet PCs and PDAs into the plant. Other radios like those based on IEEE 802.15.4 have been optimized to support wireless sensor networks for industrial instrumentation via the ISA100.11a and WirelessHART standards. Standards for wireless instrumentation and condition monitoring ISA100.11a and WirelessHART enable the creation of scalable, integrated applications based on wireless sensor networks. Wireless sensor networks based on ISA100.11a or WirelessHART deliver data to the full range of maintenance, safety, and secu-



rity applications.

In addition, reliable, long-range, high bandwidth wireless technologies like WiMAX (IEEE 802.16) have been ruggedized for industrial environments and applications. These industrial broadband networks are designed to deliver improved reliability and wireless capacity.

Managing an invisible asset

There is an extremely important asset you own and control. However, if you do not manage this asset effectively, it could turn into your greatest liability. This asset is your airwaves—the radio-frequency spectrum available to you in and around your facility. Imagine two to three years from now when you have hundreds or even thousands of wireless devices in your plant from dozens of vendors. Just like your wired networks today, without the right tools for managing the secure, effective coexistence of your airwaves, the wireless networks will become unreliable, slow, and potentially create unintended security vulnerabilities. In addition, the choices you make today may limit your options in the future as new wireless technologies become available. How do you avoid this looming wireless logjam and future-proof your wireless networks? There are three basic choices a plant can make in implementing wireless networks.



Choice 1: Select a single vendor

Choosing all of your wireless networking applications from a single vendor gives you the advantage of an engineered system designed to integrate various wireless technologies into a single seamless system. Unfortunately, this choice locks you into a limited set of lowest-common-denominator proprietary "standards" that will leave you vulnerable to being held hostage by that vendor. As new applications emerge and wireless technologies evolve, your dependence on that single vendor will limit your options and hold you back while your competitors move forward.

Choice 2: Select best-of-breed for each application need

The second option is to review the unique needs of your operations, facilities, and desired applications and choose the best wireless technology for the specific applications. This best-of-breed approach



will have a better chance of delivering the performance and reliability for that one specific solution. But each best-of-breed point solution you deploy will demand its own infrastructure and management system, requiring an investment in technology and manpower. Each point-solution, wireless technology, and application will also require its own wireless infrastructure, network management, and security approach. Each additional wireless application will be much more expensive to deploy and manage, making it difficult to establish a positive return on your investment in any single wireless technology or application.

Choice 3: Select an open, scalable wireless network

It is obvious wireless technologies are not standing still. There will be new wireless technologies, tools, devices, and applications becoming available over the next decade. Locking yourself into a single vendor or an inflexible, dedicated network and point-solution wireless system will not let you easily take advantage of new technologies as they become available.

The answer is to have a single shared wireless application network that allows plug-and-play interoperability, management, and security of any wireless devices and applications—regardless of their radio frequency, protocol, or application. A truly open wireless application network will allow you to choose exactly the right wireless device and application for your plant. This wireless application network also delivers greater application flexibility and cost certainty with an engineered approach that creates a network of systems and integrated applications based on open standards, best practices, and vendor neutrality across all your wireless applications.

The choice should be easy. The financial and operations benefits of industrial wireless are most effectively realized with Choice 3. By using the best tools for the job and an open wireless infrastructure, you will avoid the limitations of a single vendor solution and the constraints imposed by numerous point solution wireless applications. Instead, you will enjoy the full breadth of benefits of wireless in all areas of plant operations.

RESOURCES

ISA100, Wireless Systems for Automation
www.isa.org/link/isa100

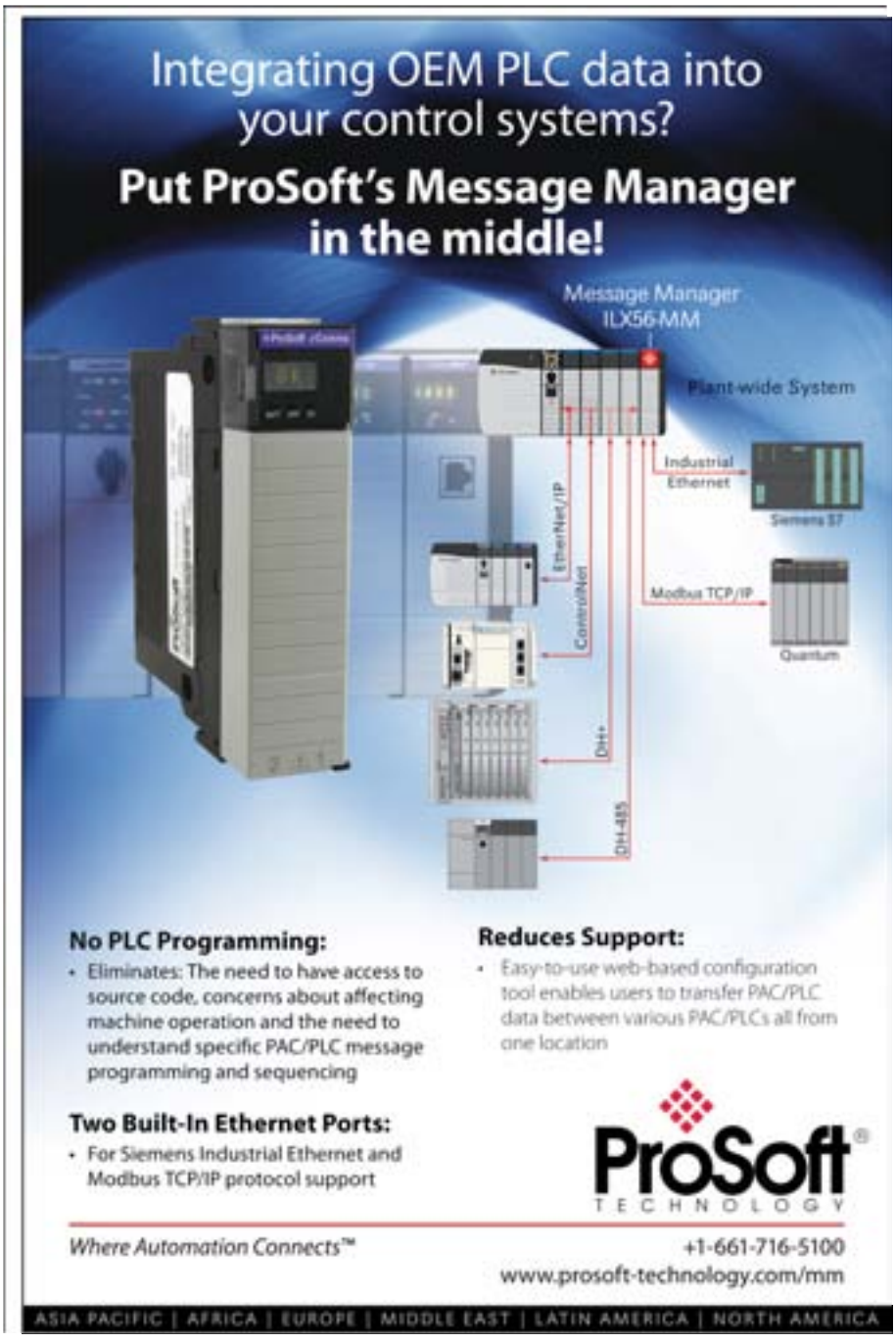
All on one network
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Wireless applications evolving
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Advances in flow and level measurements enhance process knowledge, control

By Gregory K. McMillan

Smart measurements help analyze, diagnose, predict, control, and optimize process performance



Jobs in research, process development, process design, process technology, automation, operations, or maintenance depend upon the ability to see, trend, analyze, diagnose, and collect the information from measurements. Ultimately, what you want to know as an engineer or technician is "why." Modern instruments have made great progress in not only answering "why" but also offering a higher level of plant performance. The material balance is at the core of the process and the "why." Flow and level measurements determine the material balance, and hence, process behavior.

More obvious is the performance and integrity of control systems and safety systems depend upon the accuracy, reliability, and speed of the measurements. You cannot control or protect something you cannot measure.

Modern flow and level measurement instruments have made significant advances in addressing an industrial plant's requirements through:

1. Technological advances in sensing element technology
2. Indication and integration of multiple measurements
3. Compensation of application and installation effects
4. Online device alerts and diagnostics
5. Remote configuration and calibration
6. Digital signals with extensive embedded user-selected information
7. Wireless communication

Transmitters with features 1 through 6 are classified as "smart" or "intelligent."

Since the 1980s, the out-of-the-box accuracy of modern industrial instrumentation has improved by an order of magnitude. Consider the differential pressure transmitter (DP). The 0.25% accuracy of an analog electronic DP has improved to 0.025% accuracy for a smart microprocessor-based DP. Furthermore, analog DP accuracy often deteriorated to 2% when it was moved from a nice bench-top setting to service outdoors in a nasty process with all its non-ideal effects of installation, process, and ambient conditions. A smart DP with its integrated compensation for non-ideal effects will stay close to its inherent 0.025% accuracy. Additionally, a smart DP takes 10 years to drift as much as an analog DP did in one year. (Note: Drift is an error that increases with time.)

Smart instruments offer the ability to report additional process variables, such as the local ambient and process temperature, and alerts based on abroad spectrum of device diagnostics. These variables and alerts are communicated digitally on the same signal as the primary process vari-

able that is being measured. The visibility of this wealth of additional information has been greatly improved in the modern distributed control system and the associated asset management systems. The use of the Electronic Device Description Language standard technology for the interface to smart instrumentation enables easier and more effective access to and visualization of the information. Instrument suppliers can readily provide an interface that optimizes the look and feel of the data items in their devices.

Relative accuracy of inventory measurements

Accurate flow and level measurements are needed to control material balances (control material residing in the process and entering and exiting process) and residence times (amount of time material stays in a particular unit operation). Flow and level measurements can be calibrated online by the change in inventory detected by an accurate inventory measurement.

An example of the accuracies of properly set up and calibrated inventory measurements for changing process fluid compositions are listed below where 1 is the highest and 10 is the lowest accuracy. For the level devices, random errors are most important because we are looking for a change in level that would cancel out fixed (offset) errors. Note: Even the lowest accuracy device might be perfectly adequate for many applications where knowing the exact level or change in level is not necessary, such as surge tank level. For applications where residence time and material balance control are important, such as continuous reactor and column overhead receiver level, measurement noise and sensitivity are more important than overall accuracy.

1. Radar level measurements (provided dielectric constant effects are negligible)
2. Coriolis flowmeter total (provided pipeline inventory effects are negligible)
3. Load cells (provided structural support and wind effects are negligible)
4. Ultrasonic level (provided vessel vapor effects are negligible)
5. Multivariable DP level (provided DP sensing line effects are negligible)
6. Magnetic flowmeter total (provided velocity and conductivity are above low limit)
7. Vortex flowmeter total (provided velocity and Reynolds number are within limits)
8. Multivariable DP flowmeter total (provided flow and Reynolds number are within limits)
9. Single direct connect DP level (provided DP sensing line effects are negligible)
10. Bubbler DP level (provided DP sensing line and bubbler effects are negligible)

A measurement device with higher installed accuracy can be used for the online calibration of devices with lower installed accuracy. For example, radar level and Coriolis flowmeter totals could be used to calibrate the other level and flow measurement devices on the list.

FAST FORWARD

- New sensor technologies with integrated intelligent compensation, diagnostics, and communication have made a dramatic improvement in the accuracy and maintainability of flow and level measurements.
- Level measurements are important for the inventory and control of the quantity and residence time of material essential for product quality and financials.

There are many opportunities for accurate level measurements to increase the tightness of accounting, custody transfers, batch charges, feed rates, residence times, and material balances. Some of the more prominent application examples involve tanks, columns, crystallizers, and reactors.

Storage tanks

Raw material and product storage tanks require the best level measurement accuracy, particularly resolution, sensitivity, and repeatability when used in the calculations for:

- Inventory accounting
- Custody transfer
- Batch charges
- Continuous feed rates
- Material balances

Load cells were long considered to be the best means of inventory measurement for these tanks. Since load cells incur a large engineering and installation cost for the special piping, support, and calibration system, there has been an effort to seek alternatives that offer as good or better performance on the plant floor.

Radar gauges for inventory measurements are capable of an exceptional accuracy of 0.04 inches (1.02 mm). Since radar is a surface level measurement, the accuracy in percent of span gets better as the span increases. For a 15 foot (3.8 m) tall tank, the 0.04 inch accuracy corresponds to an incredible accuracy of 0.01% of span. When used in combination with a strapping table and a Coriolis meter for density measurement in a recirculation line, a radar device is capable of an unexcelled calculated mass or volume measurement in addition to level measurement.

Columns

One of the biggest and least recognized opportunities for sensitive level measurement and tight level control is overhead receivers on distillation columns. The most widely used column strategy control is the direct material balance

scheme where temperature manipulates the distillate flow from the overhead receiver. The level in the receiver is controlled by the manipulation of reflux flow to the top of the column.

Vapor flow from the reboiler bubbles up through the column, condenses in the overhead exchanger, and accumulates in the overhead receiver. Reflux flows down the column and accumulates in the sump. A decrease in vapor flow from a decrease in steam to the reboiler or a sudden shift in wall temperature from a cold wind or cold rain will cause a decrease in overhead receiver level. For tight level measurement and control, a small change in level will quickly translate to a change in reflux flow that will balance the change in vapor flow. This inherent self-regulation provides some internal reflux control and helps decouple the energy balance from the material balance. Since reflux flow is typically higher than distillate flow (reflux/distillate ratio >1), the level controller has plenty of muscle to keep the level exactly at set point.

The performance of the temperature loop depends upon a sensitive level measurement and tight control. When the temperature loop makes a change in the distillate flow, the change in controller output has no effect on column temperature until the overhead receiver controller makes a change in the reflux flow. If the level control is tight, the correction to the column is fast. In fact, to bring the level back to set point, there is some lead-lag action because the reflux must be driven momentarily past the balance point where reflux flow equals the distillate flow to keep the level constant. What is important here is not total accuracy but the ability of the level device to consistently distinguish as small a change in the level as possible. The level measurement must have excellent resolution, sensitivity, and repeatability (precision) and negligible noise or sufficiently filterable (e.g., high frequency) noise.

Crystallizers and reactors

From the 1960s to the 1980s, load cells in weigh tanks were the way to ensure the charges to reactors were accurate. Accuracy was typically 0.1% to 0.25%. Loss in weight was used to determine the charge and to provide a mass flow rate. Today, DP level and radar level transmitters with Coriolis flowmeters provide a lower cost alternative for accurately feeding critical unit operations.

In batch and continuous crystallizers and reactors pushed to capacity, the level is operated close to its high limit. Operation above this limit is undesirable because it may be above the heat transfer area and may allow some loss of material into the overhead vent system due to foaming and swelling from bubbles, sloshing from agitation, and liquid entrainment in the vent gas.

For batch operations, a higher level offers a higher product mass per batch cycle. For continuous operations, a higher level enables a higher feed rate



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for the same residence time requirement. Conversion to product in continuous reactor vessels depends on the reactants or super-saturated components having enough time to complete the product or crystal formation. Most reaction and crystallization rates are not instantaneous. The conversion depends on residence time (inventory divided by throughput flow). Level should be kept proportional to production rate to keep the conversion constant. Excessive residence time can increase byproducts, product degradation, and crystal agglomeration.

Surge tanks

The normal purpose of a surge tank is to provide surge capacity. The level should be allowed to move up and down to absorb mismatches in process equipment operating rates upstream and downstream of the surge tank. However, when production rates are high or low, it may be desirable to operate at the tank's level limits. There may be residence time limits similar to those stated above for reactors and crystallizers to keep the product concentration and quality constant or enable the product formation rate to continue to meet production rate requirements. Operation at level or residence time limits put greater demands on level measurement accuracy.

The primary way of affecting the process is by changing a flow. Whether you are considering supervisory, model predictive control (MPC), or PID control systems, what is finally manipulated is a flow in the process industry. In production units, the final control element is typically a control valve that introduces a nonlinear gain that is the slope of the installed characteristic.

Cascade control opportunity

A flow control loop can be closed around the control valve to provide a correction of valve position to maintain a flow set point regardless of the shape of the installed valve characteristic. If the valve is manipulated by process control loops, the effect of the non-linearity and uncertainty of the installed characteristic is moved from the primary process variable (e.g., composition, level, pH, pressure, or temperature) to the secondary flow loop where it is corrected by reset.

Feedforward control opportunity

There are flow disturbances from streams going into and out of the process. If a disturbance can be measured and the effect of the disturbance on the primary process variable can be calculated or identified, the measurement of the disturbance can be used to set the manipulated flow in anticipation of the disturbance's effect on the primary process variable. The primary controller then corrects the ratio of the manipulated flow to the disturbance flow. The use of a secondary flow loop removes the nonlinearity and uncertainty of the valve characteristic from the ratio calculation. This preemptive action, taken before the feedback measurement of the primary process variable fully sees the disturbance, is called *feedforward control*.

Some examples of flow ratios for feedforward control in the process industry are:

- Coolant/feed flow ratio for crystallizer, cooler, extruder, or exothermic reactor temperature control
- Steam/feed flow ratio for distillation column, evaporator, heater, dryer, or endothermic reactor temperature control
- Distillate/feed or reflux/feed flow ratio for column temperature control
- Reagent/feed flow ratio for pH control
- Additive/feed flow ratio for blend control (e.g., percent solids)
- Air/fuel flow ratio for boiler or furnace combustion control (oxygen control)
- Feedwater/steam flow ratio for boiler drum level control (three element control)
- Blowdown/feedwater flow ratio for boiler drum total dissolved solids (conductivity control)
- Supply/demand flow ratio for header pressure control

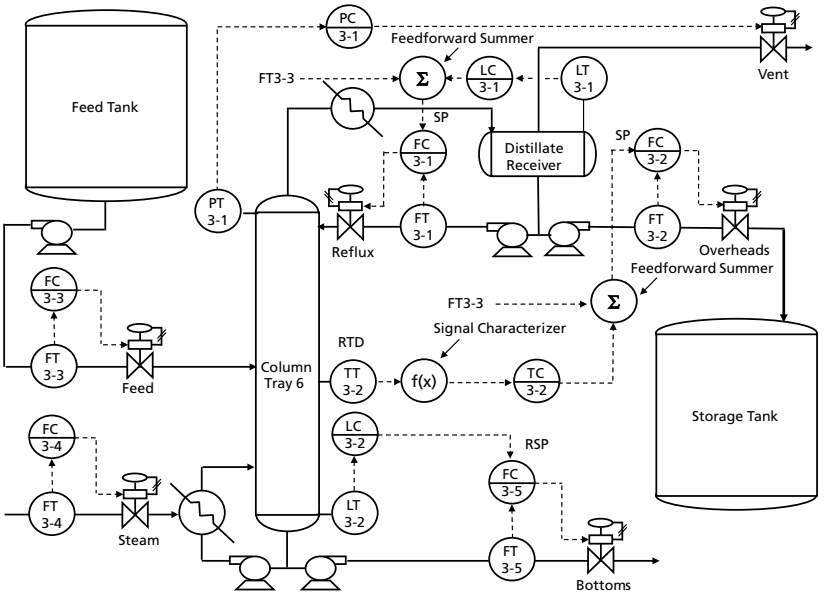
Process analysis

For billing and yield and cost-of-good calculations, accurate flow measurements are needed. As the cost-related importance of a stream increases, the need for flow measurement accuracy increases. The importance can be related to direct cost or to indirect cost from process impact.

Data analytics, such as principal component analysis (PCA), require flow measurements that are repeatable and linear. The use of valve positions instead of flow is generally unproductive because of installed valve characteristics. The use of split-ranged valves

Flow feedfoward for column temperature control

The addition of flow feedforward for column temperature control provides pre-emptive action for flow disturbances.



is even more problematic for PCA because of the discontinuous and nonlinear behavior from the additional stick-slip and flattening of the valve characteristic at the split range point.

Process modeling

Each of the following types of models benefits from accurate and repeatable flow measurements:

- Projection to latent structure or partial least squares (PLS)
- MPC
- PID adaptive controller tuning
- Neural network
- First principle

Flows determine what is going on in a process. If you do not get the flows right, not much else matters. Because of valve backlash, stick-slip, nonlinearities, and variable pressure drop, all types of process models have suffered from the use of valve positions rather than flow measurements. PLS, MPC, and PID models assume dynamics that are linear and independent of direction and size—all bad assumptions when valve positions rather than flows are used as inputs. Additionally, the valve nonlinearity from the installed characteristic varies with pressures at the inlet and outlet of the valve. Even first principle models with pressure-flow solvers to compute pressures have not fared well because of the uncertainties in piping resistances and valve responses. Additionally, parallel pressure-flow solvers have exhibited numerical instabilities resulting in simulation crashes for the extreme operating conditions that occur during batch operations and the startup and the activation of safety instrumentation systems in continuous processes.

Pioneering advances in dynamic modeling offer a next generation of pressure-flow solvers in dynamic models that will be robust and adapted enough to provide flows from valve positions. The ability to consistently and comprehensively compute flows for all streams will enable these models to reach the highest levels of fidelity required for research, development, and design of automation systems. Presently, models can only move up in fidelity when flow control loops are installed on the key streams so feedback action removes the nonlinearity and unknowns of the valve and piping system. New pressure-flow solvers can eliminate this precondition. A side benefit will be the demonstration by these models of the process control improvement that can be gained from cascade, feedforward, and ratio control. The quantifiable benefits from demonstrable test cases can justify new flow devices to provide missing flow measurements or improve the accuracy of existing flow measurements.

RESOURCES

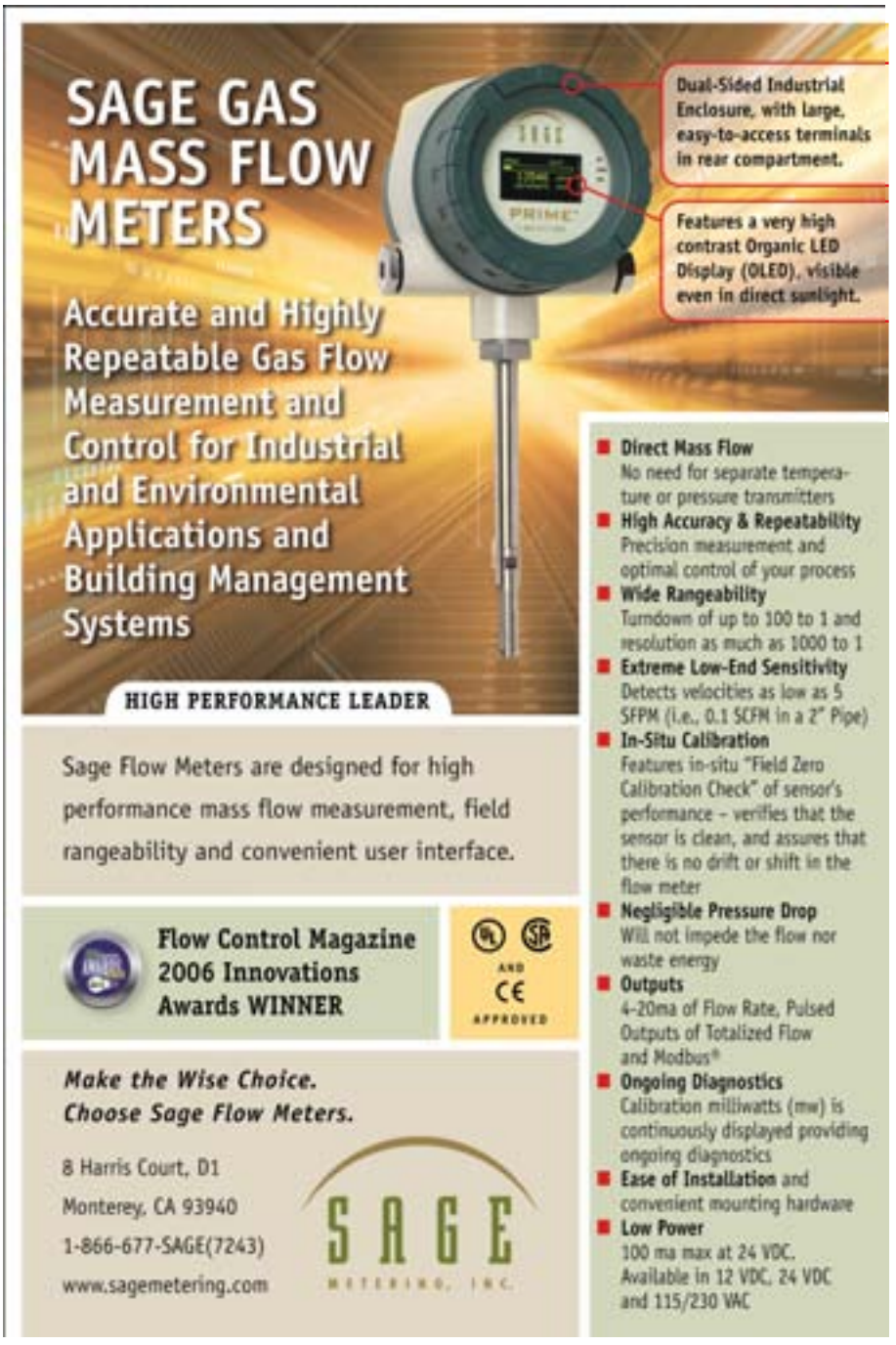
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Source: *Essentials of Modern Measurements and Final Elements in the Process Industry: A Guide to Design, Configuration, Installation, and Maintenance* by Gregory K. McMillan (ISBN: 978-1-936007-23-3) www.isa.org/finalelements

View the online version at www.isa.org/intech/20100204.



Clamp-on ultrasonic flowmeter improvements

Ultrasonic flow measurement technology offers a low-cost method to measure flow. The advantage of clamp-on ultrasonic flow sensors is installation without stopping a process to put a hole in a pipe to insert a conventional sensor.

By John Erskine, Michael Scoon, and Brian Sternberg

FAST FORWARD

- Non-invasive installation saves installation cost and time.
- Newer technology is more reliable and accurate.
- Improved accuracy with higher sample rates.

Two years ago, at a flowmeter presentation for a state-owned oil company in the Middle East, we led off with a tried-and-true technology: Differential pressure meters. As expected, this technology was well-known and fully accepted throughout the organization. As we went down the list of technologies in our portfolio, clamp-on ultrasonic meters met with a cooler reception: "No disrespect intended, but frankly we've tried everybody's, and we could not get any to work …"

History

Since their introduction, clamp-on ultrasonic flowmeters have had a shaky reputation in the field of flow measurement.

Original models relying on the Doppler principle were often misapplied, as the promise of a non-invasive solution exceeded the limits of the technology. (Doppler meters were never suitable for clean liquids.) Experienced field people tell stories of installing the first ultrasonic meters near metropolitan areas, where the meter tended to behave more like an AM radio.

In the 1990s, transit time ("time of flight") ultrasonic technology became widespread—a more robust solution was now available for clean liquid applications. However, this too was far from a panacea: Depending upon the make and model, users were still limited as to the levels of turbulence, straight pipe requirements, minimum fluid velocities, and overall accuracy at which the flowmeter could perform successfully. On the whole, despite relatively lower costs for installation and maintenance, clamp-on ultrasonic flowmeters have remained a second (or third) choice for process engineers around the world.

Technology development

Ultrasonic flowmeter manufacturers have not sat idly by during this time; you may be surprised to learn just how far clamp-on ultrasonic technology has come. Today's clamp-on ultrasonic meter is highly advanced, making the technology well worth a fresh look by engineers and field





A set of clamp-on ultrasonic transducers mounted on a chilled liquid system.

instrumentation managers. Major improvements include data acquisition speed, speed of sound measurement, and digital communications.

As microprocessor technology continues to improve, so do the capabilities of the ultrasonic flowmeter. As late as five years ago, most ultrasonic meters were making raw readings at a rate of less than 10 times per second. Today, this rate is poised to exceed 100 times per second. Such a vast improvement allows for substantial enhancements to performance in two critical areas—response time to changes in flow rate and more sophisticated data filtering. Manufacturers are leveraging these improvements into more accurate (and more repeatable) flowmeters that respond almost instantly to varying flow rates.

Transit time flowmeter accuracy is a function of fluid speed of sound measurement, and in this area manufacturers also have made major strides. Engineers have refined techniques to measure speed of sound on the fly, improving signal-to-noise ratios and on the whole pushing the envelope at which transit time technology can succeed. Improvements to sound signal transmission ("signal strength") further this cause, and today's transit time meters work on applications where they would have failed years ago—dirtier or gaseous fluids, thicker pipe walls, larger pipes, and applications with greater amounts of turbulence.

Over the past 15 years, transit time metering has continued to evolve, and now it includes multi-path technology that employs multiple wetted sensors. The sensor array, which can range from three to six paired sensors, takes several readings across the fluid stream (liquid or gas) and produces accuracies that meet government weights and measures standards. Multipath ultrasonic flowmeters are widely used in custody transfer processes where line sizes exceed 4 inches, replacing Coriolis and differential pressure flowmeters, which have been the leading technology for many years. Although these are in-line meters, they deserve special mention here to underscore the position that ultrasonic technology in general has secured.

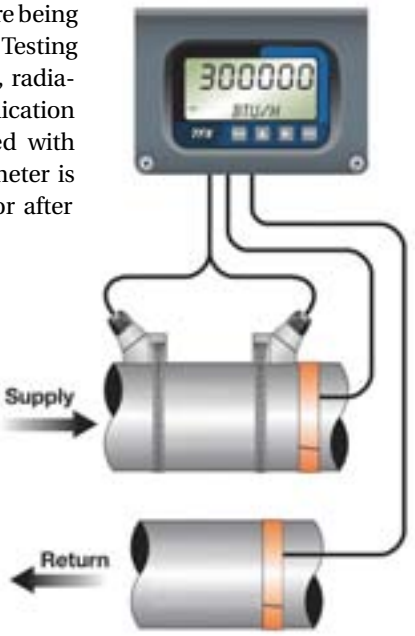
Transit time liquid flowmeters have benefited from advanced microprocessor technology, in terms of greater accuracy and greater success rates out of the box.

Industrial communications

Finally, as with other flowmeter technologies, today's clamp-on ultrasonic meters feature a host of communication and software enhancements to make them compatible with advanced control systems: Digital communications such as Modbus and Ethernet IP; heat energy measurement calculations along with industry standard communications such as BACnet; and user-friendly meter programming and configuration via USB.

New applications

Clamp-on ultrasonic meters today are being utilized in new applications as well. Testing and monitoring of heat exchangers, radiators, and chillers is a growing application for ultrasonic flowmeters combined with energy calculation software. The meter is clamped around the pipes before or after the heat exchanger; high-accuracy RTDs (also available as non-invasive devices) are attached at the inlet and outlet. By measuring the flow and differential temperature, users can calculate the energy removed in BTUs [$(Q^*\Delta T)/500$]. Equipment efficiencies can be calculated as a baseline, and ongoing performance can be continuously monitored. When using a clamp-on ultrasonic meter over a mechanical technology, one need not be concerned with pressure loss, contamination, or impeding flows through the system. As engineers and operators become more concerned with managing energy and resources, ultrasonic meters become a strong choice to support these efforts.



An ultrasonic flowmeter utilized in energy flow application. One set of clamp-on ultrasonic transducers is mounted on the supply or return line; a clamp-on Pt1000 RTD is mounted on the supply and return lines. Measuring volumetric flows and the difference between supply and return temperature allows for calculations of heat energy usage.

There is always room for improvement, but clamp-on ultrasonic flowmeters have truly begun to deliver on their original promise with greater accuracy and reliability. Use of ultrasonic flowmeters has increased due to improvements and simple non-invasive installation.

Ultrasonic meters are worth new consideration for flow measurements.

ABOUT THE AUTHORS

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View the online version at www.isa.org/intech/20100205.

Applying Coriolis technology to high pressure applications



Measurement at high pressure needs heavy-duty metering

By John Daly

The first commercially available Coriolis flowmeter based upon the observed effects of mass flowing through vibrating tube systems appeared in the late 1970s, early 1980s. Any mass flowing through such tubes causes dampening and distortion effects in the tubing system, which correlates to the actual flow in the piping. Today, many Coriolis mass flowmeters in use employ this technology, with about a dozen different manufacturers producing them. All of these meters are built on a vibrating piping system with the inertia of the mass of material flowing through creating very small but measurable deflections of the tubing.

The name for these meters comes from the force responsible for the deflections—the Coriolis force. Compared to other technologies, which mostly determine flow velocity, Coriolis mass flowmeters offer direct mass flow measurement, and unlike velocity measurement techniques, changes in density, viscosity, and flow profile do not, in general, play a significant role when measuring flow with a Coriolis meter.

These tremendous advantages, along with a noninvasive nature, lack of moving parts (which together equate to a high turndown ratio and minimal maintenance), and high intrinsic accuracy (typically 0.15–0.5%) have made Coriolis meters very desirable as measuring elements. Coriolis meters are traditionally higher priced than many other metering technologies; but over the last few years, prices have dropped due to designs oriented more to bulk manufacturing techniques and resulting economy of scale. Manufacturers have responded to the requirements of specific industries by offering dedicated solutions to applications such as natural gas dispensing, and this has led to further adoption of Coriolis meters in wider market areas.

Excitation force applies

Coriolis mass flowmeters have different designs with regard to shaping the tube system for measuring the flow. The U shape was initially one of the most popular tube geometry designs, and it serves to illustrate the theory of Coriolis flowmeter functionality.

The principle of operation is this: Application of an excitation force to the U-shaped tubes causes them to oscillate backward and forward, while flow enters the tubes at one side of the meter, travels through the tubes, and exits at the other side.

The oscillation of the tubes is orthogonal to the material flowing within them. As material passes through the tubes, the flowing mass accelerates in the direction of the oscillation. Due to its inertia, the tubing sees a force—the Coriolis force—that adds to the deflection of the tube around the oscillation axis. The tube form takes on a double-bended or "S" shape. This additional bending registers as a phase shift and is directly proportional to the mass passing through the tubes.

The additional bending, which is most pronounced in the middle of the U shape, is a direct result of Coriolis force and relates only to the mass moving through the meter. The more

mass flow, the stronger the Coriolis force and the more pronounced the bending. The range of bending is very small, typically in the range of one ten-thousandth of a millimeter up to a few tenths of a millimeter, depending on design. For accurate, stable mass flow measurement, a good signal-to-noise ratio is necessary—high signal means significant bending and deformation of the tubes while low noise requires external factors like vibrations to contribute to the primary measured deformation. Both factors strongly relate to the design of a Coriolis flowmeter.

The design of a meter influences the strength of the induced Coriolis force by its geometric layout and the amplitude/frequency of oscillation. To achieve larger tube deformation and therefore more precision in measurement, fast and energetic oscillation and a long "arm" (the distance from the upper U shape to the oscillation axis) is desirable. The momentum responsible for the deformation is proportional to these parameters. By contrast, working against the deformation by the Coriolis force is the elastic module of the tubes—here called the spring constant. The higher the spring constant, the less deformation that registers at a particular Coriolis force. Considering this, the design goals for a Coriolis flowmeter should not only consider the creation of a large Coriolis force, but also the spring constant of the tubes themselves, as this can work against large deformations/deflections.

Considering these interdependencies, at first glance the ideal Coriolis mass flowmeter should have the following features:

- Long, easily bendable tubing systems
- Large distance between oscillation axis and excitation point
- Thin wall tubing to keep the spring constant small
- High energy excitation to create large oscillation amplitudes

Unfortunately, some practical aspects of meter design temper the design goals for the ideal meter:

- Long piping systems may create unacceptable pressure drop.
- High-pressure, abrasive, and/or corrosive media require thick wall tubing.
- High energy input may conflict with safety requirements in hazardous areas.
- Excessive excitation may lead to fatigue failure of the tubing.

The above, while strongly simplified, shows there is no one ideal Coriolis mass flowmeter design. Regardless of where design emphasis lies, there will always be tradeoffs made for applicability and practical suitability against performance.

It is worth noting the performance of a Coriolis flowmeter can suffer from external noise, and installation quality remains a key factor in Coriolis flowmeter performance. While installing the meter in a vibration-free environment can minimize external noise issues, the design of the meter itself must be optimal to prevent problems. Here the criterion is to keep the system oscillation stable and difficult to disturb from the outside. It is important that balanced tubing is part of the hardware so the swinging is very stable and possibly self-sustaining—like a tuning fork—and the actual point of deformation measurement is well decoupled from any process connection stress and influence.

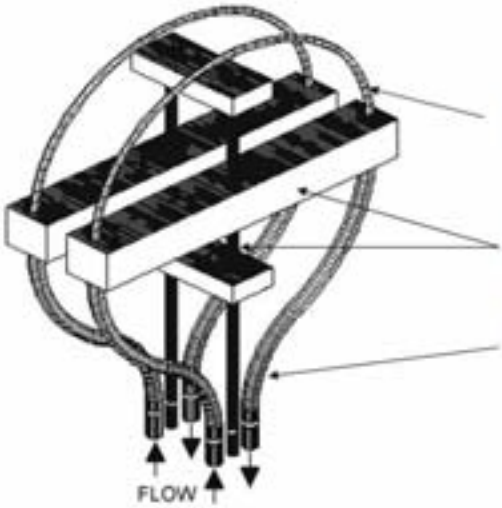
Over time, the various Coriolis meter designs have gotten better and are now nearly all suitable for the "run-of-the-mill" applications found in everyday processing. The differences and limitations of these designs only become fully visible when exposing the meters to extreme applications and conditions, and it is in these applications that the omega tube torsion Coriolis mass flowmeter design demonstrates some very distinct advantages over other designs.

FAST FORWARD

- Semi-circular measuring element is unaffected by pressure change.
- Oscillation system supports meter tubes giving resilience to external vibration interference.
- ANSI Class 900, 1500, and 2500 flanges are not uncommon on omega tube design meters.

Omega tube meter features

The omega tube torsion Coriolis mass flowmeter is unique in that it is equipped with torsion rods and crossbars as part of its mechanism. This design has proved to be universal, serving "standard" mass flow applications and, more important, extreme applications with high flow



Design features of omega tube meter

Decouples half-circle measurement section

Carefully balanced tors bar oscillation system

Process feed tubes

rates, pressures, and/or temperatures:

- Rates up to 1500 m/hour
- Line sizes up to 12"
- Pressures up to 900 bar (13,000 psi)
- Temperatures from –250°C (–482°F) to 400°C (752°F)

The omega tube design is different from traditional Coriolis meters. The active part of most Coriolis mass flowmeters consists of its oscillating tubing, while the omega tube system consists of three distinctly different mechanical elements, each dedicated to an essential function of the flowmeter:

- **Half-circle measuring tube**—The Coriolis force deforms this part, and it is therefore the active measurement element in the meter. Installed here are deformation sensors in the form of pickup coils and magnets.
- **Torsion bar oscillation system**—This system consists of two torsion rods and two crossbars providing the base oscillation system. It would oscillate even without tubing attached.
- **Process feed tubes**—This section is below the mass bars and sees almost no bending from the meter oscillation; it sees only a low-stress torsion moment.

The separation of the functional elements gives the opportunity to optimize each element separately according to its function and mitigate the tradeoffs between the "ideal" design in the sense of creating strong measurable signal and application requirements, such as heavier wall thickness and wetted material selection.

Three-inch omega design meter with ANSI Class 2500 flanges



The active measurement element is fixed left and right where it passes through the crossbar. Since the section is an exact semicircle, pressure changes—even the highest ones—do not alter its shape. Other designs have a natural tendency to slightly lose the shape of their original form when pressurized and this affects accuracy. This means an omega tube meter calibrated on water at low pressure, for example, can serve to measure compressed natural gas in the field at a pressure between 200 and 300 bar (2800 and 4200 psi) without affecting precision or zero.

External noise effects remain at a minimum because the measurement section is isolated from the majority of such noise by the mass of the crossbar and, apart from the pick-up devices that measure the deflection induced by the Coriolis force, the section is free of additional elements that could resonate and disturb the measurement.

The real key to the omega tube flowmeter's suitability to difficult applications is the unique oscillation system. The system's characteristic components are the torsion rods and the crossbars. Each torsion rod with its crossbar represents an oscillation system on its own—it is like a tuning fork that works independently, even without attached tubing. Excitation energy for oscillation is injected via coils sitting on the crossbars themselves. The torsion rod serves by storing this injected energy, smoothly delivering it into the oscillation movement as the tubes swing back and forth. The use of crossbars in conjunction with torsion rods creates very energetic and stable oscillation with very little energy input and once oscillating at harmonic frequency, the meter is very mechanically tolerant of disturbing and dampening effects because of the mass of the crossbars. A standalone tubing system does not build up and keep this amount of oscillation energy. The energy requirement of an omega tube meter to sustain oscillation is in fact so low that even full 6" piping with a wall thickness of over 5 mm can still be rated intrinsically safe for zone 0. Furthermore, the large amplitude of oscillation generated is not critical in terms of mechanical stress to the tubing system. Unlike in a classical U-shape configuration, a large movement in the active section only results in small noncritical torsional movement in the process feed tube section and not in an "over-bending" of a tube. This feature makes it possible to use heavy wall thickness tubing while still allowing the generation of large amplitude oscillations.

For most Coriolis mass flowmeters, the tubing actually represents the spring constant, and hence changes in the tubing directly influence the operation. Tubing in the omega tube design plays a secondary role with regard to the spring constant of the meter, and changes in the tubing type can easily be accommodated without major re-engineering, allowing the use of very different materials from standard stainless steel to exotic (with very different elasticity module) materials like Tantalum.

Extreme diameter-to-wall thickness ratios are not a problem either. For instance, tubes with an outside diameter of 114 mm and a wall thickness of 11 mm work without degrading performance. The ability to design meters with "standard" wall thickness tubes rather than thin wall tubes also means much higher pressure applications can be metered—ANSI Class 900, 1500, and 2500 flanges are not uncommon on omega tube meters—along with the welcome advantage of not having to fit the necessary secondary pressure housings needed to provide containment in the event of a tube failure. The use of thick wall tube and pipe within the omega tube meter translates into high confidence in safety when installed on site.

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View the online version at www.isa.org/intech/20100206.

RESOURCES

The two phases of Coriolis flow
www.isa.org/link/Jan2009Coriolis

Coriolis: Twist and Shout
www.isa.org/link/200306Coriolis

Flow Research, Inc.
www.flowresearch.com

Online collaboration: A win for all of us



By Tom Moser

There was a time, not long ago, that organizations gauged customer needs via controlled market research, input this feedback to the marketing group, who then worked with engineering to conceptualize new products and develop new offerings to fulfill unmet needs. The process was understood, fairly linear, and certainly controlled by manufacturer or provider. Success of course was directly correlated to how well the control group represented the whole.

But the days of gathering customer feedback from such a traditional process are now numbered, if not gone altogether. 'Focus Groups' or 'Advisory Boards' will remain important, but are being complemented by online forums, communities, blogs, and other forms of online social media.

Today, as consumers, we have the opportunity to evaluate, share, research, and comment on any product or service online. Consumers can change the course of a new product introduction and influence what companies will develop and sell to us in the future. While this might put us outside of our traditional comfort zone, one thing is for certain—we all need to accept it and embrace it. In the article, "Time's Person of the Year in 2006: You," Lev Grossman said, "The new Web is a very different thing. It's a tool for bringing together the small contributions of millions of people and making them matter."

But can we capture online activity and channel it to influence—and improve—new product ideas and, in turn, even better address the needs of customers? All signals point to yes. A recent Aberdeen Group report titled "The ROI on Social Media Monitoring: Why it Pays to Listen to Online Conversation" surveyed 250 companies directly engaged in social media and identified that those organizations realized a 93% improvement in their ability to capture consumer insight that drove a new product or service development. In addition to contributions to new product development, these organizations achieved an estimated 63% customer service cost reduction and 82% improvement in identifying and reducing risk to their brands.

So how does our industry succeed in this environment? How do we capture the on-the-ground knowledge and demand of customers? How do we allow public collaboration to ensure success for all involved? We encourage all our employees to get engaged and involved online … to make a difference. In the world of Yahoo! Groups, it is estimated 1% of the user population starts a group, 10% actively participate and may author content, but 100% of users benefit from the activity. Resources required for social media monitoring are fairly small, but a commitment to regularly tracking and using information is critical. For developing an online presence or peer-to-peer network, the resources are obviously much greater. Online endeavors take dedicated resources to manage and monitor such sites. Another facet not always apparent is securing the actual adoption and involvement of internal employees. The rewards are worth it in order to strengthen collaboration with users and harness the value and insights only customers can offer.

Users of equipment certainly have different challenges: How can I ask questions online and learn from others if my processes are proprietary? As a specialist in a particular area, where do I go to find a community with interests like mine? If I spend time contributing, what's in it for me? The benefits to the participants are many:

- Direct connection to people with answers to your questions
- The ability to tap ideas/solutions from virtually anywhere
- Sharing of best practices
- Discussion of common issues and unmet needs
- Access to the latest information on new products and services
- In-depth training, without leaving your office
- Personal development via accelerated learning
- Opportunity to be part of a global network of peers with common interests

Social media is dramatically changing our behavior as end consumers. In the B2B world, it is time to fully leverage the capabilities that Web 2.0 enables. We will all win.

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Magnetic flowmeter technology

Magnetic flowmeters utilize Faraday's Law of Electromagnetic Induction to determine the velocity of a liquid flowing in a pipe. Faraday's Law forms the basis for electrical generation systems where wires travel through a magnetic field and produce a voltage.

In a typical physics class experiment to illustrate the phenomenon, a wire (conductor) connected across a galvanometer can be moved through the magnetic field of a horseshoe magnet and cause the galvanometer pointer to move. Moving the wire in the opposite direction will cause the pointer to move in the opposite direction due to the changing voltage polarity. Moving the wire faster will cause more voltage to be generated and the movement to move higher.

In magnetic flowmeters, a magnetic field is generated and channeled into the liquid flowing through the pipe. To accomplish this, the electromagnetic coils can be located outside of the pipe (flow tube), however the flow tube must be non-magnetic to allow penetration of the magnetic field into the liquid. Locating the coils internal to the flowmeter (closer to the liquid) can reduce the electrical power necessary to deliver the magnetic field, as well as reduce the size of the flowmeter and fabrication costs.

Following Faraday's Law, flow of a conductive liquid through the magnetic field will cause a voltage signal to be generated. This signal is sensed with electrodes located on the flow tube walls. When the coils are located externally, a non-conductive liner is installed inside the flow tube to electrically isolate the electrodes and prevent the signal from being shorted. For similar reasons, non-conductive materials are used to isolate the electrodes for internal coil designs.

The fluid itself is the conductor that will move (flow) through the magnetic field and generate a voltage signal at the electrodes. When the fluid moves faster, more voltage is generated. Faraday's Law states the voltage generated is proportional to the movement of the flowing liquid. The transmitter processes the voltage signal to determine liquid flow.

Electromagnetic excitation

The voltage signal will take the same general form as its electromagnetic excitation. When a magnetic flowmeter is excited by a sinusoidal magnetic field (AC waveform), the signal generated at the electrodes is also sinusoidal. In earlier designs, these signals were subject to a number of influences that affected measurement quality, including stray voltages in the process liquid, capacitive coupling between the signal and power circuits, capacitive coupling between interconnecting wiring, electrochemical voltage potential between the electrode and the process fluid, and inductive coupling of the magnets within the flowmeter. These flowmeters required a zero adjustment to compensate for these influences and the effect of electrode coating.

Turning the electromagnetic field on and off (DC waveform) causes the signal to resemble a square wave. When the electromagnetic field is on, the signal due to flow plus noise is measured. When the electromagnetic field is off, the signal due to only noise is measured. Subtracting these measurements cancels the effects of noise and eliminates the zero adjustment, reducing the abovementioned drift problems and improving performance.

Waveforms other than those described above are also in use.

Measured quantity

The magnetic flowmeter signal is proportional to the fluid velocity. Therefore, these flowmeters measure velocity, from which the volumetric flow rate is inferred utilizing the first equation, assuming the cross-sectional area of the conduit is known. Magnetic flowmeter performance is therefore predicated on how well the average fluid velocity is measured and how well the cross-sectional area is known. Uncertainty in the cross-sectional area can degrade the inferred volumetric flow rate measurement.

Typical applications

The construction of the magnetic flowmeter is such that the only wetted parts are the liner and electrodes, both of which can be made from materials that can withstand corrosion. In addition, the straight-through (obstructionless) nature of the design reduces the loss of hydraulic energy across the flowmeter (pressure drop) and the potential for abrasion from the flowing liquid. Therefore, magnetic flowmeters can measure many corrosive liquids and abrasive slurries.

Magnetic flowmeter liners and electrodes can be constructed of materials that do not contaminate the liquid. Therefore, these flowmeters can be applied when liquid contamination is an issue, such as in sanitary applications.

Straight run requirements are relatively short, so magnetic flowmeter technology can be applied where limited straight run is available. In addition, magnetic flowmeter technology has no Reynolds number constraints, so it can be applied where the liquid exhibits high or varying viscosity.

Magnetic flowmeters that sense velocity and level can measure the flow of liquids in partially filled pipes, such as interceptor sewers and storm water culverts. Magnetic flowmeters with fast response times can measure liquids that flow for relatively short periods of time, such as in batch and fill operations.

Magnetic flowmeters measure liquid velocity, from which the volumetric flow rate is inferred. The measurement is linear with liquid velocity and exhibits a relatively large turndown.

Source: *The Consumer Guide to Magnetic Flowmeters, 2nd Edition* by David W. Spitzer and Walt Boyes, Copperhill and Pointer, Inc., 2004.

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Industrial Flow Measurement, 3rd Edition, ISA, 2005.

Flow Measurement, 2nd edition, ISA, 2001.

Signal conditioners: The 'ins' and 'outs'

By Jay DeCastro

Whether you call them signal conditioners, isolators, converters, or interfaces, these useful process instruments solve important ground loop challenges every day.

The need for signal isolation began to flourish in the 1960s and continues today. Electronic transmitters were quickly replacing their pneumatic predecessors because of cost, installation, maintenance, and performance advantages. However, it was soon discovered that when 4-20mA (or other direct current, or DC) signal wires have paths to ground at both ends of the loop, problems are likely to occur.

The loop in question may be as simple as a differential pressure transmitter sending a 4-20mA measurement to a receiver, such as a recorder. But when the voltages (V) at the two ground points are different, a circulating, closed current (I) path is formed by the copper wires used for the 4-20mA signal and the ground. When this happens, an additional and unpredictable amount of current is introduced into the loop, which distorts the true measurement. This current path, known as a ground loop, is a very common source of signal inaccuracies.

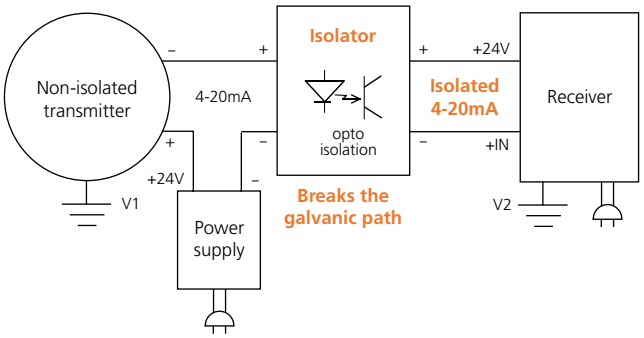
A ground loop forms when three conditions are present:

1. There are two grounds.
2. The grounds are at different potentials.
3. There is a galvanic path between the grounds.

To remove the ground loop, any one of these three conditions must be eliminated. The challenge is the first and second conditions are not plausible candidates for elimination. Why? Because you cannot always control the number of grounds, and it is often impossible to just "lift" a ground.

The ground may be required for the safe operation of an electronic device. It is also possible the ground exists because the instrument is in physical contact with the process which, in turn, is in physical contact with the ground. From a practical standpoint, you cannot reach into the earth and regulate the voltage at these permanent ground points.

So what can be done? Use a signal isolator to "break" the galvanic path between the two grounds. When the conductive path between the differential voltages is broken, a current cannot form. The ground loop has been eliminated.



A signal isolator "breaks" the galvanic path between two grounds.

Breaking the galvanic path

The first and foremost duty of an isolator is to break the galvanic path between circuits that are tied or "grounded" to different potentials. A galvanic path is a path in which there is a direct electrical connection between two or more electrical circuits that allow current to flow. Breaking this galvanic path can be accomplished by any number of means. For most industrial measuring equipment, the two prevalent methods chosen for galvanic isolation are optical and transformer isolation.

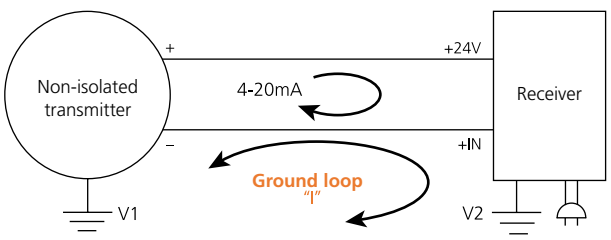
Optical isolation uses light to transfer a signal between elements of a circuit. The opto-coupler or opto-isolator is usually self-contained in a small compact module that can be easily mounted on a circuit board.

An optical isolation circuit is comprised of two basic parts—a light source (usually a light emitting diode (LED), acting as the transmitter) and a photo-sensitive detector (usually a phototransistor, acting as the receiver). The output signal of the opto-coupler is proportional to the light intensity of the source. The insulating air gap between the LED and the phototransistor serves as the galvanic separation between circuits, providing the desired isolation between two circuits at different potentials.

Optical isolation has better common-mode noise rejection, is usually seen in digital circuits, is not frequency sensitive, is smaller, and can sometimes provide higher levels of isolation than transformer isolation.

Transformer isolation, often referred to as electromagnetic isolation, uses a transformer to electromagnetically couple the desired signal across an air gap or non-conductive isolation gap. The electromagnetic field intensity is proportional to the input signal applied to the transformer. Transformers are very efficient and fast at transferring alternating current (AC) signals. Since many process control signals are DC, they must be electrically "chopped" into an AC signal so they can pass across the transformer. Once passed, they have to be rectified and amplified back into the desired DC signal output.

The ability to depend on accurate monitoring and control signals is literally priceless. Signal conditioners enhance measurement accuracy and protect signals from damaging conditions, thereby saving money.



A ground loop forms when the voltages at two ground points in a loop are at different potentials.

ABOUT THE AUTHOR

Jay DeCastro is a project engineer at Moore Industries-International, Inc.

Thriving during the economic downturn by building a real-time enterprise

By Peter G. Martin

Part 1 of a two-part series

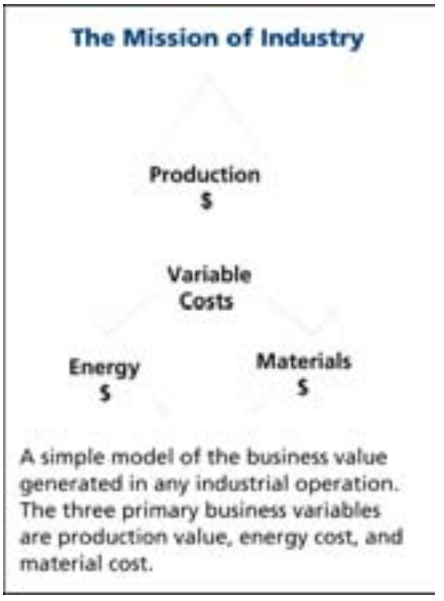
Industrial companies commonly respond to a difficult economy by hunkering down and waiting for the market to swing back up. Although this is often a safe approach, it sometimes results in lost opportunities to increase production, improve efficiencies, and better the overall operations of the plant. As the economy weakens, capital spending is naturally cut, which is a very reasonable reaction. But when capital projects stop, new opportunities arise to use existing technology and talent to drive performance improvements. By seizing these opportunities, not only will the company cope with the impact of a weak economy, it will also be in a strong position to capitalize on new efficiencies when the economy recovers.

Industrial plants are actually a collection of assets working together to create or generate products. These assets include process assets, such as equipment and piping; automation and information assets, such as instrumentation and systems; supply assets, such as energy and feedstock; and human assets, such as operators, maintenance personnel, engineers, and management. To optimize the overall business performance of the enterprise, each of these asset groups must operate collaboratively at peak levels. Because existing human assets are not busily applied to new capital projects during an economic slowdown, this might be the ideal time to divert human asset capabilities to fundamentally enhance the performance of all the other assets.

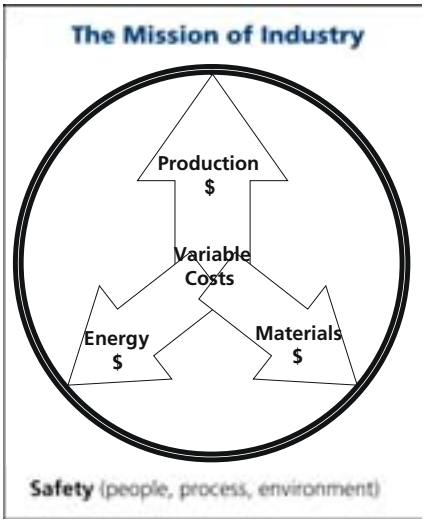
While this might be obvious to industrial management, what might not be as obvious is determining an effective strategy the plant's human assets can employ to drive performance improvement. Identifying the appropriate strategy is most readily accomplished by evaluating the business dynamics that have changed over the past few years. To determine what might be new and different in the industry, it might be useful to start at the business basics of the plant.

At their core, industrial plants exist to maximize production value by generating variable products at the lowest possible cost. Traditionally, process automation has primarily been used to reduce the fixed cost of industrial operations—typically in the form of headcount reductions. After decades of headcount reductions, there are so few people left in most industrial operations that there is just no room left for reductions. Since fixed cost reductions are no longer viable, industrial companies must now start to focus on reductions in the variable costs of the operations. In the process industries, the majority of these variable costs come from energy and materials.

The good news is most industrial operations already have the tools to move to a real-time business operations management strategy.



A simple model of the business value generated in any industrial operation. The three primary business variables are production value, energy cost, and material cost.



When evaluating the three primary business variables for dynamic changes that have occurred over the past few years, we find important shifts have occurred. For example, only a few years ago, industrial companies could negotiate energy contracts with their suppliers over a period of six months or even one year, effectively making the price of energy a constant value. Today, long-term contracts are no longer available in most markets and, not only that, the price of energy might change many times every day. This has had a domino effect on the other key business variables to the point which today all three key business variables of industrial operations are starting to move toward real-time variability. This shift has completely altered the business dynamic and created the need for an effective business management system. Since their inception, manufacturing processes have had to operate in real time. Now, the business of manufacturing has to be conducted in real time too.

As a result of these dynamic shifts, traditional strategies for managing manufacturing operations fall short. Most of the business management infrastructure in industrial operations has been designed and implemented to provide monthly

snapshots of businesses that today are changing minute by minute. This is unacceptable. It is tantamount to driving a car while only looking through the windshield once a month, hoping you arrive to your desired destination without an accident. It is just not going to happen.

Exacerbating this situation is the freedom to drive improved business value is limited by mandatory safety programs that focus on protecting the people, the process, and the environment. In many cases, these safety standards impose constraints on business variables and often work at odds with business objectives.

Many of variables that measure safety fluctuate in real time as the plant dynamics vary. For example, the process and equipment measurements that foreshadow an impending accident or event, such as equipment vibration or temperature, change in real time with equipment operation. Even environmental issues, such as emissions, can change in real time. These safety variables must be managed in real time while simultaneously driving real-time business improvements even as the other key business variables are also varying in real time. It is obvious the most effective strategy to improve performance in such a real-time atmosphere is to develop a real-time business operations management infrastructure that is built to match the dynamics of emerging real-time enterprises.

It is important to note, many industrial business executives are under great internal, public, and regulatory pressure to make their operations environmentally sustainable, which is part of the overall safety variable. Environmental sustainability encompasses emissions, energy, effluent, water, and other key variables that can be impacted by an industrial operation.

The good news is most industrial operations already have the tools to move to a real-time business operations management strategy. They just might not be the tools most business managers look to when addressing other business issues. The second part of this article, coming in March/April *InTech*, will show how to measure the business in real time, provide empowerment for plant operations employees, and delineate the relationship between real-time operations business management and real-time profit optimization.

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Peter G. Martin Ph.D. is vice president for business value generation at Invensys Operations Management. Martin has spent three decades in the automation industry, culminating with the development of commercially-applied dynamic performance measurement technologies and methodologies. An established author and industry speaker, he received the ISA Life Achievement Award in 2009 for his work in performance measurement.



ISA99: Charting a security standards roadmap into a risky new decade

By Charley Robinson

Not that we needed the reminder, but the near-miss bombing of Delta/Northwest flight 253 in the waning days of 2009 underscored what we all know: Murderous forces in the world seek vulnerabilities wherever they can find them. Those who work in industrial automation can tell you many manufacturing and processing operations can also present such opportunities for those who know what they are doing.

And so as ISA Standards enters its seventh decade, a major focus continues on standards and guidelines to reduce the possibilities and limit the impacts of cyber threats to industrial systems and critical infrastructure. This work is performed primarily by the ISA99 committee on Industrial Automation and Control Systems Security, but draws from and impacts work across the ISA standards world and beyond.

The ISA99 roadmap

In the past year, ISA99 has established a roadmap that calls for delivering at least 14 standards and technical reports addressing vital aspects of industrial control systems security. These documents will build on ANSI/ISA-99.00.01, *Security for Industrial Automation and Control Systems: Concepts, Terminology and Models*. That standard was followed by ANSI/ISA-99.02.01, *Security for Industrial Automation and Control Systems: Establishing an Industrial Automation and Control Systems Security Program.*

Work is underway on the roadmap in areas including system security requirements and security assurance levels, target security assurance levels for zones and conduits, compliance metrics, and patch management. Also in progress is an update of a key 2007 technical report, ANSI/ISA-TR99.00.01, *Security Technologies for Manufacturing and Control Systems.* It provides an assessment of cyber security tools, mitigation countermeasures, and technologies that may be applied to industrial automation and control systems regulating and monitoring numerous industries and critical infrastructures.

In support of the roadmap, ISA99 has established three new working groups on cyber security with other key ISA standards committees. These include joint working groups with ISA100 on wireless automation, ISA67 on the special requirements of nuclear plants, and ISA84 on functional safety.

The joint working group with ISA84 highlights the importance of understanding the impact of cyber security on the safe operation of industrial processes. As technologies such as wireless, Ethernet, and computer information systems gain increased usage in industrial automation, the need for design strategies and methodologies to identify and mitigate risk is clear.

The joint working group leverages the expertise found in the ISA84 and the ISA99 committees to address these challenges. The leading work of ISA84 in functional safety has been a foundation of the widely used International Electrotechnical Commission (IEC) standards on safety in the process industries. "The ISA84 work, and subsequent work in IEC 61508 and IEC 61511, identifies cyber security as a potential threat to safe operation, but the scope of ISA84 focuses mostly on hardware faults and device reliability," said ISA84 Chair William Johnson of DuPont. "The ISA99 joint working group with ISA84 represents a significant complement to our work, as it addresses faults and emerging threats today that jeopardize safe operations in ways that many were less concerned about even just a few years ago."

The ISA99-84 joint working group's initial work has focused on:

- Developing a security assurance level methodology for cyber security, similar to that of the current safety integrity levels (SIL) defined in ISA84
- Defining and developing processes for identifying intentional and systematic threats that can expose process hazards

"Today when we consider only the probability of hardware failures in a hazards analysis, we can miss significant sources of risk to process safety," said ISA99 co-chair Eric Cosman, engineering solutions architect with The Dow Chemical Company. "This can be a dangerous assumption in the modern interconnected and software-driven plant, when considering intentional threats such as viruses, malware, and hackers, but also unintentional systematic faults such as poor network performance or network failures."

Another roadmap

The ISA99 work has also been recognized within the *Framework and Roadmap for Smart Grid Interoperability Standards* released by the U.S. National Institute of Standards and Technology (NIST) in September 2009. NIST's intent is to identify existing and draft standards vital to the success of the highly publicized Smart Grid program. All ISA99 published and draft documents are being made readily available for access by U.S. state utility commissions, the Federal Energy Regulatory Commission, and the National Association of Regulatory Utility Commissioners, who will be reviewing the content of all identified standards for regulatory purposes.

ABOUT THE AUTHOR

Charley Robinson (crobinson@isa.org) is manager of Standards and Technology at ISA.

RESOURCES

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www.isa.org/standards

U.S., regional, and international standards
www.nssn.org

International Electrotechnical Commission standards
www.iec.ch

International Organization for Standardization standards
www.iso.ch

$250 million effort for science, math teachers



The White House announced a $250 million public-private effort in January to improve science and mathematics instruction, aiming to help the nation compete in key fields with global economic rivals.

With funding from high-tech businesses, universities, and foundations, the initiative seeks to prepare more than 10,000 new math and science school teachers over five years and provide on-the-job training for an additional 100,000 in science, technology, engineering and math, according to *The Washington Post*.

The initiative effectively doubles, to more than $500 million, a philanthropic campaign for so-called STEM education that President Barack Obama launched in November 2009. Separately, the government spends about $700 million a year on elementary and secondary education in the STEM fields through agencies such as NASA, the National Science Foundation, and the U.S. Education Department.

Intel Corp., based in Santa Clara, Calif., and the Intel Foundation are committing $200 million in cash and in-kind support over 10 years for expanded teacher training and other measures. For instance, the company will offer nationwide an intensive 80-hour math course to help U.S. elementary school teachers, who are usually generalists, develop expertise.

The Woodrow Wilson National Fellowship Foundation, a nonprofit organization in Princeton, N.J., will expand a program that places math and science teachers with advanced degrees in hard-to-staff schools in Indiana, Michigan, and Ohio. With $40 million in foundation and state funding, the program will train 700 teachers over three years.

Other elements of the initiative include a $13.5 million expansion of a university-based program called UTeach that aims to deliver 7,000 expert teachers by 2018; a commitment from public universities to prepare 10,000 math and science teachers a year, up from 7,500 annually; and efforts by NASA and PBS to promote effective math and science teaching.

"If we're going to be economically competitive and continue to innovate and create jobs, we have to get much, much better in STEM education," said Education Secretary Arne Duncan. "There's a huge sense of urgency."

Chinese curb power use

Some Chinese factories were ordered last month to shut down to ensure sufficient power to heat homes as demand surged amid record-setting winter cold, a utility company said.

According to *The Associated Press,* no outages were reported, but coal supplies were running low at power plants in central China, said Liu Xinfang, a spokesman for State Grid Corp., which operates most of China's power-distribution network.

"Power demand is greatly increased because people need to stay warm," Liu said. "Our facilities are in excellent shape, but we lack coal. It's like cooking without rice."

Temperatures in Beijing plunged to 3°F (-16°C) recently, the lowest in 33 years, the China News Service reported.

Last winter, some areas of China suffered blackouts after power companies let coal stockpiles run low due to high costs and snowstorms disrupted supply efforts. China relies on coal for about 70% of its power.

In Hubei province in central China, some factories were ordered to shut down because power demand outstripped supply, said Liu. He said State Grid was moving power to Hubei from other provinces.

"We are putting a priority on residential power consumption," Liu said. "We are asking factories to take turns cutting power use. Then we will ask commercial facilities to limit power use."

Japanese project aims to turn CO_2 into natural gas


Source: energydaily.com

Japanese researchers said they hoped to enlist bacteria in the fight against global warming to transform carbon dioxide (CO_2) buried under the seabed into natural gas.

The researchers at the Japan Agency for Marine-Earth Science and Technology aim to activate bacteria found naturally in earth to turn CO_2 into methane, a major component of natural gas, reports *AFP.*

A team led by chief researcher Fumio Inagaki have already confirmed the bacteria exists in the crust deep under the seabed off the northern tip of Japan's main island, a spokesperson said.

But the project faces a big challenge to develop a method of activating the bacteria and accelerating the speed of methane gas generation, a spokesman for the agency acknowledged.

In the natural environment, the bacteria turn CO_2 into methane gas very slowly, over billions of years, he said. The researchers hope to develop technology within about five years to activate the bacteria and shorten the transformation time to about 100 years, he said.

The aim is for the bacteria to produce methane gas from CO_2 buried in a layer about 6,600 feet under the sea bed, the agency said.

Changing price paradigms

By Jim Pinto

This "Channel Chat" series will cover a wide variety of industrial instrumentation and automation topics. We started with a brief history of automation (www.isa.org/intech/Channel_201001) and will branch out into innovation, standards, manufacturing, systems integration, sales channels, and different kinds of success in this business. Wherever my nose points, and your feedback leads …

Today's world has three business/technology models:

- U.S. businesses develop products with 60 to 70% gross profit margins, and target revenue growth of $100 million to $1 billion. U.S. investment is simply not available for products with smaller markets and margins. Because of this, U.S.-developed products are more complex and are targeted for large markets that can justify higher investments.
- Developing countries (other than China) are growing rapidly through products that have intermediate complexity with smaller revenue growth and medium (40 to 50%) gross-profit margins. In India, Brazil, and other developing countries, there are exciting technology companies growing to $5-10 million within three to five years with medium complexity products, quickly developed. This level of success attracts relatively high levels of investment.
- China is unique in that target gross-profit margins are only 5 to 10%, margins considered too small anywhere else. It is this that has made China the world leader in low-price manufacturing of high-volume products.

It must be emphasized that the profit margins being discussed here are gross-profit, the manufacturing cost related to net selling price, and not net-profit, after sales, development, and administrative expenses are accounted for.

In the U.S., gross-profit margins of about 60% typically result in a target net pre-tax profit of 15-20%. In many other countries, a gross-profit of 40% results in net profit of 5-10%, which is considered acceptable. In China, gross-profit is typically 10-15%, and the net-profit is typically less than 2%. That is the essence of global pricing differentials.

Automation suppliers endeavor to maximize profit margins by emphasizing proprietary products, with design features that can command higher margins. But the global, fast-moving technology treadmill quickly demolishes that lead; few high-volume products cannot be quickly copied.

Pricing alternatives

Conventional cost-based pricing is stuck in a trap. Products manufactured offshore at a lower cost are not the answer—not just because the manufactured cost may be lower, but because global companies are prepared to compete with lower profit margins.

The tactical response by the large automation suppliers is to offer a broad range of products, software, systems, and services. But this still has the effect of reducing overall profit margins. My contention is the problem lies in the obsolescent concept of cost-based pricing.

In today's changing global markets, no other marketing decision highlights the double-edged conflict/cooperation nature of the buyer-seller relationship. Pricing is a zero-sum game in which one's gain is the other's loss. The focus must move to a win-win business relationship, simultaneously providing greater customer value and higher supplier profitability.

It is useful to consider the risk/reward trade-off embodied in various pricing approaches. Typical fixed-price, cost-based sales involve only cost risk for the seller. The price is set before the product or service is made or provided. Pricing for services based on costs plus a predetermined profit margin involves no vendor cost risk. The customer pays for all cost overruns, and the supplier's profit is established before delivery.

Especially for large systems and integration services, performance-based pricing is the answer. The seller is paid based on actual performance of products and services.

Performance-based pricing is "insurance" the seller does not undercharge the buyer; it guarantees that as the seller provides more, it is paid more. Significantly, the buyer also receives insurance it will not overpay; it pays only for the amount of performance actually delivered on a measurable basis.

Of course, this means the performance and expected results of the product must be immediately measurable. With the availability of machine-to-machine (M2M) communications, the system results can be monitored consistently to provide the required performance measurements.

Performance-based pricing must include installation, service, and maintenance because performance is attained only when the product or system is operating. In return, the seller should expect to achieve a high return based on performance.

For example, a $100,000 system typically entails a prolonged budgetary/purchasing procedure. Performance-based pricing can be structured to simplify and speed up the process, providing the buyer with a relatively low front-end cost barrier. The contract can be structured to break-even in less than a year, provided the expected performance is achieved, with further incentives for the supplier to exceed financial results. Of course, the supplier must afford the front-end cash-flow; this is typically not a problem for larger companies.

Performance-based pricing moves the cost and price risk to the seller. Neither is established before the deal is made. But the supplier then gets the opportunity to manage the value to the customer and be closely involved to generate additional profits for both sides. With the risk comes added revenue and profit opportunity.

In today's competitive global business environment, traditional cost-based pricing is seriously flawed. Performance-based pricing should be examined as a viable alternative.

ABOUT THE AUTHOR

Jim Pinto is an industry analyst and founder of Action Instruments. You can e-mail him at jim@jimpinto.com or view his writings at www.JimPinto.com. Read the Table of Contents of his book, *Pinto's Points,* at www.jimpinto.com/writings/points.html.

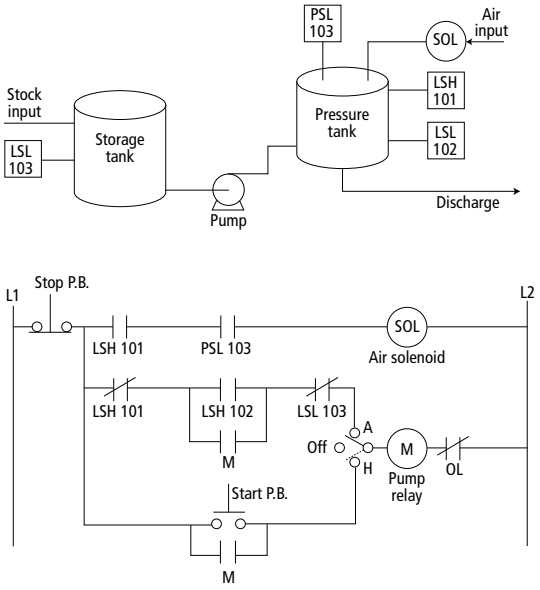
Documenting skills is value-add

ISA certification provides an objective, third-party assessment, and confirmation of a person's skills. It gives manufacturing and factory staff the opportunity to differentiate themselves from their peers and gain recognition. *InTech* covers two certification areas in this monthly Certification department.

ISA Certified Automation Professional (CAP) program

CAP question

Given the process shown below, with the relay logic shown, which of the following statements is true?



A. When the pump control switch is in auto, and the pump is stopped, a low pressure in the pressure tank will start the pump.
B. When the pump control switch is in auto, and the pump is running, a low pressure in the pressure tank will stop the pump.
C. When the pump control switch is in auto, and the pump is running, a low level in the pressure tank will stop the pump.
D. When the pump control switch is in auto, and the pump is running, a low level in the storage tank will stop the pump.

CAP answer

The correct answer is **D**. The storage tank low level switch LSL103 or the pressure tank high level switch LSH101 can stop the pump.

The pressure tank low pressure switch PSL103 is not in the motor control circuit, so a high pressure or low pressure does not have an impact on the pump status. The pressure storage tank low level switch LSL102 is involved in starting the pump, but does not stop the pump since the motor seal-in circuit is in parallel.

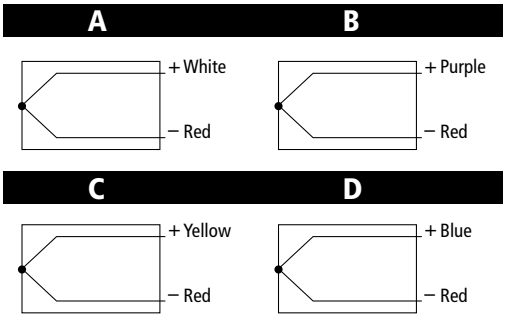
References: Bryon Lewis, CSE, P.E.; *Control Systems Engineering Exam Reference Manual: A Practical Study Guide,* ISA Press, 2007. Thomas A. Hughes; *Programmable Controllers, 4th Edition,* ISA Press, 2005.

ISA Certified Control Systems Technician (CCST) program

Certified Control System Technicians (CCSTs) calibrate, document, troubleshoot, and repair/replace instrumentation for systems that measure and control level, temperature, pressure, flow, and other process variables.

CCST question

Which of the following is the standard wiring color for a Type K thermocouple in the U.S. and Canada?



CCST answer

The correct answer is **C,** which illustrates the very common Type K, Chromel/Alumel thermocouple.

A illustrates the very common Type J, Iron/Constantan thermocouple.

B illustrates the less common Type E, Chromel/Constantan thermocouple.

D illustrates the less common Type T, Copper/Constantan thermocouple.

References: Bryon Lewis, CSE, P.E.; *Control Systems Engineering Exam Reference Manual: A Practical Study Guide,* ISA Press, 2007. Thomas A. Hughes; *Measurement and Control Basics, 4th Edition,* ISA Press, 2007.

Focus on signal conditioning

Remote I/O systems



The LB/FB Remote I/O Systems are designed to facilitate the connection of traditional input and output signals onto a bus network, LB/FB systems support general-purpose or intrinsically safe applications with modules that can be mixed and matched on the same backplane. These systems facilitate connection to PROFIBUS DP, Modbus, and Foundation fieldbus for all traditional signal types, including discrete, analog, frequency and temperature. Redundancy for the communication interface and power supply ensures high reliability.

LB/FB users benefit from: Reduced wiring; fewer DCS control cards; fewer terminations for reduced installation costs; reduced control cabinet space needs; and faster commissioning.

Pepperl+Fuchs, www.pepperl-fuchs.com



Transducer



The MINI Analog Shunt Transducer is ideal for converting and isolating mV signals from shunt resistors. The new transducer features three-way 1.5 kV isolation, low power consumption, and high-quality signal transmission.

Current shunt resistors are precision resistors used in the measurement of DC electrical currents by creating a voltage drop across the resistor. The MINI Analog Shunt Transducer then filters, isolates, and converts this mV signal to a standard analog current or voltage signal. The transducer can be configured to accept -50mV to 3V drops. The signal output is also configurable to commonly used analog signals.

The MINI Analog family has a 6.2mm thin housing and T-Bus power bus capabilities that save space and drastically reduce installation time. The hot-swappable signal conditioning slices simply clip onto the powering T-Bus connector without removing bus bars or bridging.

Phoenix Contact
www.phoenixcontact.com

Universal signal conditioners

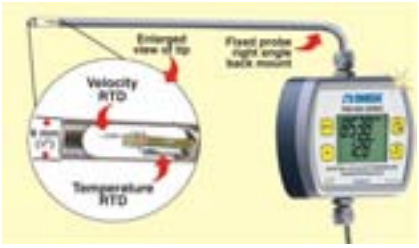


Universal signal conditioners, in plastic slim-line housings, convert, isolate, and transmit scale signals from a wide variety of process sensor and controller I/O. The DIN-rail mountable models (884114 and 84116) support scalable input signals including mA, VDC, thermocouple with internal cold junction compensation, two to four-wire RTDs, linear resistance, and potentiometer signals. Both models feature mA and VDC outputs, while the 84116 model adds two individually programmable relays for alarming and control functions. Isolated universal supply voltage input eliminates the need for separate transformers or power supplies.

A menu-structured LCD programming/display module, sold separately, features automatic scrolling text identifying each menu item. The detachable module can store and transfer configuration parameters from one signal conditioner to another, minimizing set up time with multiple unit applications. The module supports seven programming languages and can be password-protected to prevent unauthorized configuration changes. When not used for configuration, the display module can be used to display input signal values, engineering units, output signal, and relay status.

AutomationDirect
www.automationdirect.com

Air velocity/temperature transmitter/indicator



The FMA1000 series measures and displays air velocity & air temperature of air flows in ducts, pipes. This product is used in HVAC applications, research labs, and other manufacturing processes. The FMA1000 series offers many standard features such as back-lit LCD display of air velocity and temperature (can be displayed in different Engineering Units), two analog outputs corresponding to air velocity and temperature, high and low velocity voltage alarm outputs, USB serial interface, and a Windows based PC interface software. The sensor probe is available in three different configurations, fixed top mount, right angle mount, and remote probe.

OMEGA Engineering
www.omega.com

Thermal mass flowmeter



The ST75 flowmeter delivers direct mass flow measurement of gases in an easy to install insertion style instrument that requires almost no maintenance over a long life. The ST75 is ideal for the measurement of CO2 in a variety of industrial applications. The ST75 is designed for demanding process industry plant environments. With an extensive set of standard features, the ST75 is designed for line sizes from 0.25 to 2.0 inches (6 to 51mm). The ST75 provides three unique outputs: the mass flow rate, totalized flow and media temperature.

Fluid Components International
www.fluidcomponents.com

Integrated smart camera



The BOA vision system is an integrated smart camera that comprises all of the elements of an industrial machine vision system. Powerful and quick-to-deploy, the BOA is ideal for automated quality inspection applications and factory automation. BOA, an all-in-one machine vision solution, is the first smart camera in its class to incorporate multiple processing engines. This enables algorithm optimization via DSP, application management via CPU, and sensor management via FPGA. It is the first smart camera that offers truly embedded application software, which is easily set-up through a standard web browser.

DALSA Corporation
www.dalsa.com



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The engineer will live and work in Saudi Arabia and will report to the manager of process control at the plant site. Cutler Technology is the prime contractor for implementation of ADMC at the plant site and will utilize engineers from the site.

Please send resumes to jimmy.cutler@Cutler-Tech.com. The type engineers we are seeking are ones who are young in age or young in spirit. We are looking for engineers who get excited by working on the frontier of a new technology. A BS degree in Chemical Engineering is the minimal requirement. Experience building or using predictive controllers is a plus.

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The Analyzer Specialist will live and work in Saudi Arabia and will report to the manager of process control at the plant site. Cutler Technology is the prime contractor for implementation of ADMC at the plant site and will utilize personnel from the site.

Please send resumes to jimmy.cutler@Cutler-Tech.com. The type of personnel we are seeking are ones who are young in age or young in spirit.

ISA

Sample of Jobs Available at ISAJobs.org

See more at ISAJobs.org, where you can search for available jobs or advertise positions available within your company. ISA Members post resumes at no charge.

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Leveraging predictive maintenance to achieve greener field operations

By Jim Fererro



Announcements in the energy industry about efforts to go "green" are often met with skepticism and resistance, largely due to fears of high cost or implementation difficulty.

However, even basic steps, such as changing how field personnel operate and integrating equipment into existing SCADA systems, can produce efficiencies that directly benefit the bottom line.

Evolving from the "milk run" model

Most oil and gas fields are traditionally operated under a "milk run" model, meaning a field technician visits sites based on a schedule, or a daily milk run. Under this model, if a compressor shuts down after the site visit, it could potentially go unnoticed until the next scheduled visit, meaning up to 24 hours of downtime and lost production and revenue.

In an industry where downtime can represent huge financial losses, being proactive, not reactive, has a significant impact on the bottom line, and a company's carbon footprint.

For example: Take a compressor that has a discharge temperature shutdown at 300°F. During the technician's typical "milk run" visit, the compressor's panel shows a discharge temperature of 265°F. The technician moves on since there appears to be no issue with the unit.

What the technician does not know is the temperature should only be 235°F, and it has been rising. If it were monitored remotely, the temperature would be measured constantly in real time from a central control room. The operator would be well aware of the increase because of the trending capabilities of the SCADA system and could dispatch a technician as the trend was identified. The technician would arrive at the scene with knowledge of the issue and, most likely, would have the right part with him to quickly remedy the situation.

This ability to proactively address field problems, avoid shutdowns, and reduce miles driven each day saves hours of time and expense. It also makes a positive impact environmentally and has a significant effect on safety statistics, as most reportable health, safety, and environment incidents are vehicle and driver related.

Automating field operations

Reducing unnecessary site visits via remote monitoring is important when working in environmentally sensitive areas, especially when considering impact to vegetation and disturbed wildlife.

To illustrate this point, a major water project in Wyoming was producing significant amounts of precipitation that had to be disposed of or harvested as a product. Previously, the company sent trucks to pick up the water, creating unnecessary traffic. By integrating remote PLCs into the company's SCADA system, the company automated the water capture and piped it directly to disposal facilities. This field automation has reduced traffic, decreasing the company's environmental impact.

Emissions are strictly regulated by the Environmental Protection Agency (EPA) and states. If a company acquires new fields or wells, new equipment has to meet EPA and state requirements for NO_x, CO, and NMHC limits. Remote monitoring of certain equipment can ensure measurements are integrated into SCADA systems for recordkeeping. If existing regulations are tightened, more accurate records and monitoring will be mandated, and remote capture of data will be the only realistic way to comply with EPA standards.

Remote monitoring also reduces impact from spills and unplanned hydrocarbon releases. In a SCADA-monitored field, operators can immediately dispatch staff to control the accident, significantly reducing environmental impact and making cleanup easier and faster.

In an industry where downtime can represent huge financial losses, being proactive, not reactive, has a significant impact on the bottom line, and a company's carbon footprint.

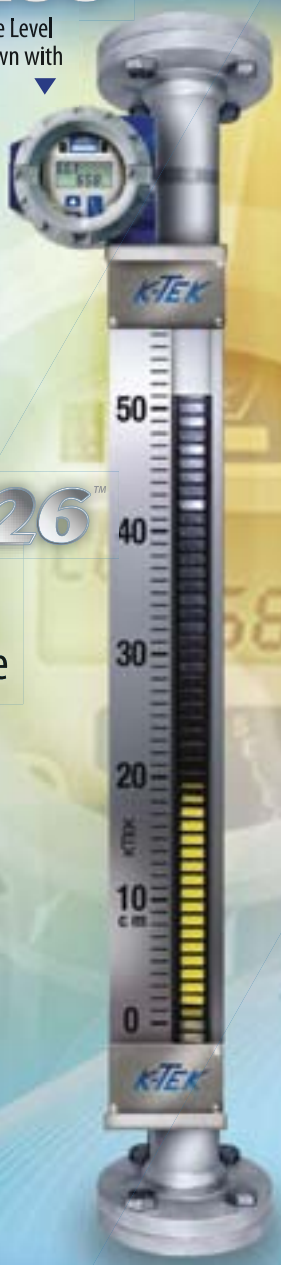
ABOUT THE AUTHOR

Jim Fererro is a founder and vice president of GlobaLogix, a Houston-based oil field automation company that helps oil and gas companies achieve greater efficiency, productivity and accuracy in their oilfield operations by providing access not just to data, but to the right information at the right time. For more information, visit www.globlx.com.

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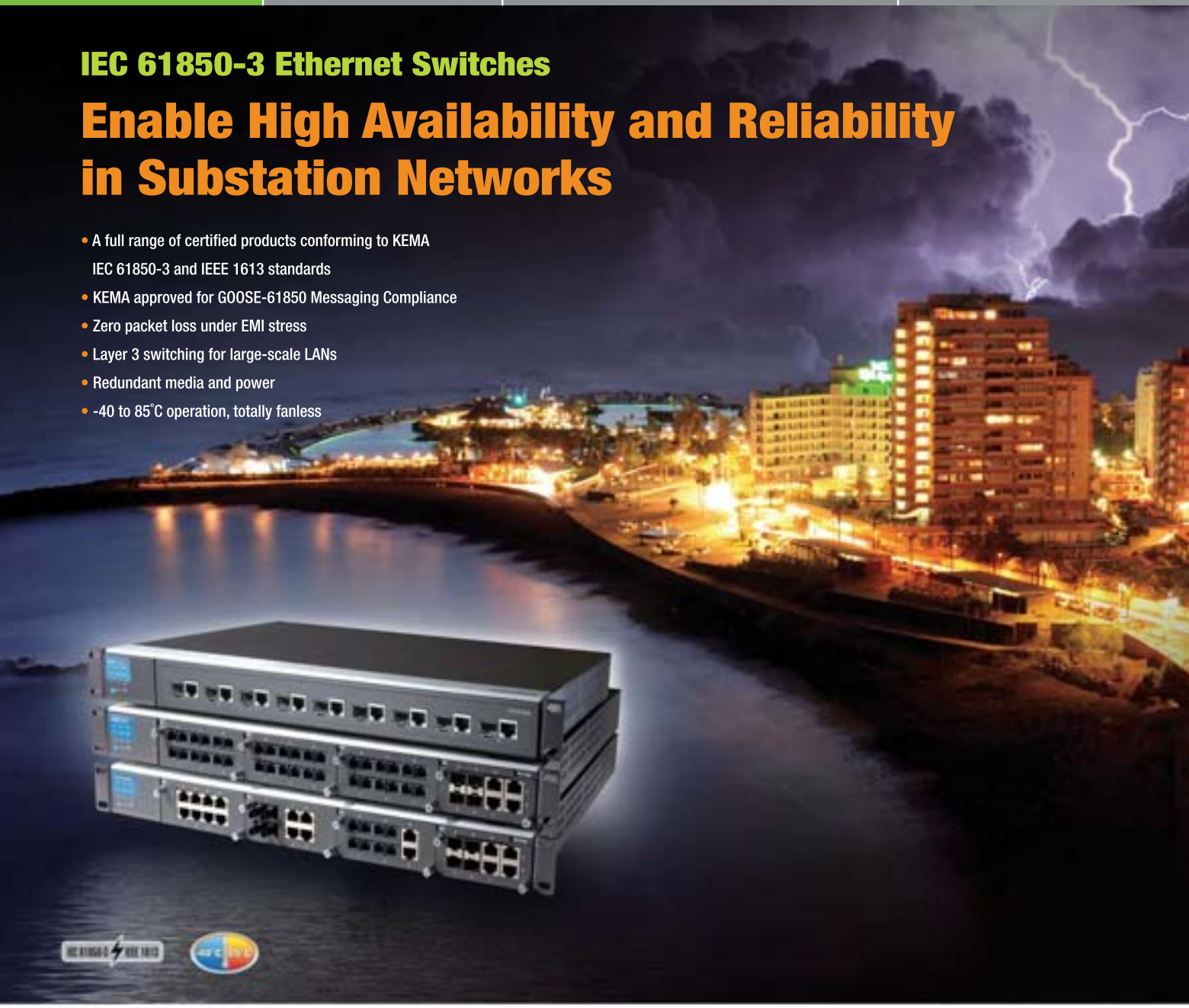
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